Affective Mirage? Diagnosing Visual Grounding in Emotion Recognition with Vision Language Models Supplementary Material

Anonymous FG2027 submission
Paper ID ****

TABLE OF CONTENTS

A Detailed Experimental Setting 2
A.1 Detailed Setting for Diagnosis 1 (General Mirage) . . . 2
A.2 Detailed Setting for Diagnosis 2 (Causal Mirage I) . . . 2
A.2.1 Paired Inputs Definition . . 2
A.2.2 Prediction-Change and Distributional Summaries . 2
A.2.3 Control Contrasts Setting for Diagnosis 2 3
A.3 Detailed Setting for Diagnosis 3 (Causal Mirage II) . . . 3
A.3.1 Cross-Sample Pairing Rule 3
A.3.2 Evaluation Anchor and Paired Records 4
A.3.3 Control Contrasts Setting for D3 4
A.4 Detailed Setting for Diagnosis 4 (Certainty Mirage) . . . 4

B Further Discussion 5
B.1 Discussion: Why AffectGPT Behaves as a Negative Control . . . 5
B.2 Discussion of prediction entropy (D4 model vs human uncertainty) . . . 5

C Additional Results for General Mirage 5
C.1 Open-ended prompt . . . 5
C.2 Unrelated natural image . . . 6
C.3 Label-order permutation . . . 6
C.4 Abstention option . . . 6
C.5 Summary . . . 7

D Additional Recognition Results for Causal Mirage 7
D.1 Diagnosis 2: Causal Mirage in Subject Removal . . . 7
D.2 Diagnosis 3: Causal Mirage in Subject–Scene Mismatch . . . 8
D.3 Summary . . . 8

E Controlled Analyses for Causal Mirage 9
E.1 Diagnosis 2: Subject-Removal Controls 9
E.2 Diagnosis 3: Subject–Scene Mismatch Controls . . . 9
E.3 Mask Appearance and Resizing . . . 11
E.4 Emotion Labels and Subject–Scene Replacement . . . 12
E.5 Prompt and Subject-View Sensitivity Analysis . . . 12

F Internal States Analysis using Activation Patching 15
F.1 Protocol . . . 15
F.2 Choice of decoder depth . . . 15
F.3 Restoration from Copied Visual-Token States . . . 15
F.4 Prediction restoration maps . . . 16

G Implementation Setting 17
G.1 Pseudo Bounding Box Generation for CAER . . . 17
G.1.1 Pseudo Subject Bounding Box . . . 17
G.1.2 Pseudo Face/Head Bounding Box . . . 17
G.2 Prompt for Diagnosis 1 . . . 18
G.3 Prompt for Diagnosis 2 . . . 18
G.4 Prompt for Diagnosis 3 . . . 19
G.5 Prompt for Diagnosis 4 . . . 19
G.6 Full Prompts for Sensitivity Analysis 19

A. Detailed Experimental Setting

A.1 Detailed Setting for Diagnosis 1 (General Mirage)

D1 uses the RAF-DB test split (3,068 facial images). All predictions are mapped to the canonical seven-class taxonomy $\mathcal{Y} = \{\text{neutral, happy, sad, anger, surprise, disgust, fear}\}$. A fixed free-text–to–canonical map resolves surface variants (e.g., the prompt option word *angry* maps to *anger*); a non-empty output matching none of the seven maps to *other*, and an output carrying no `[[·]]` span is a parsing failure. Under this dataset, *neutral* accounts for a 0.222 fraction, which is the accuracy a model attains by always predicting the dominant label; we use 0.222 throughout as the no-vision collapse anchor.

Each face is queried under four conditions that progressively remove visual evidence: (i) **image present**: the original face; (ii) **blank**: a fixed 224×224 mid-gray image; (iii) **noise**: a Gaussian-noise image generated deterministically from the run seed; (iv) **blank+mirage-hint**: the blank image together with misleading metadata stating that an image is attached. For open models, absence is instantiated as blank/noise inputs; for closed-source models we additionally send no image at all (true-absent), which replaces the hint condition in the multi-run results.

The basic prompt and its abstention-option (AO) variant (verbatim in Appendix G.2) are used for all models. Under AO, a response counts as an abstention only if the selected option is *cannot determine*; every other output, including parsing failures, is a mirage. AO is applied to the three no-vision inputs (open models: S=50 seeds; closed-source models: 50 independent calls) and to the full test set for *image_present* (C.4). Each response is classified into exactly one of three mutually exclusive types: *valid emotion* (a parseable label in $\mathcal{Y}$, further split into *collapse-to-neutral* vs. *off-neutral*), *abstention* (the model explicitly signals that the visual evidence is unavailable), or *parsing failure* (malformed / no label span).

We use stochastic decoding with temperature = 1.0, `max_new_tokens`= 64, and thinking disabled for all models. Each random seed jointly determines the Gaussian-noise realization (when applicable) and the token-sampling pattern, so a "seed" is a single noise×sampling draw.

Under a no-vision condition, an ideal evidence-aware model abstains; every response other than an abstention counts as a mirage, whether a valid emotion prediction, an out-of-taxonomy label, or a parsing failure, since none of them handles the missing visual evidence. Over inference runs the *Mirage Rate* is the proportion of runs that do not abstain:

$$\begin{aligned} R_{\text{mirage}} &= \Pr\big[\text{response} \neq \text{abstention}\big] \\ &= 1 - \Pr\big[\text{abstention}\big]. \end{aligned} \tag{1}$$

To probe stability under stochastic variation, each no-vision condition is additionally repeated for S=50 independent seeds, and the *Collapse Rate* toward the fixed-seed dominant label y_{collapse} (*neutral* for most models) is

$$R_{\text{collapse}} = \frac{1}{S}\sum_{s=1}^{S} \mathbb{1}[\hat{y}_s = y_{\text{collapse}}], \qquad S = 50. \tag{2}$$

R_{mirage} measures whether the model fails to recognize the absence of evidence, whereas R_{collapse} measures the stochastic stability of the collapse. For a no-vision condition, the input is identical across all items within a seed and the RNG is reset per seed, so per-item outputs within a seed are not independent; the independent unit is the seed. We therefore report single-seed (seed 0) and multi-seed (mean over S=50) values as a dual entry (Table I and Table S5), and estimate seed-level rates with Wilson score intervals.

We evaluate seven models spanning three families: closed-source GPT-5.6-Luna and Gemini-3.5-Flash-Lite (via a hosted API; output heads inaccessible), open-source general Qwen3.5-9B, InternVL3.5-8B, and Ministral-3-8B, and emotion-specialized Emotion-LLaMA and AffectGPT.

A.2 Detailed Setting for Diagnosis 2 (Causal Mirage I)

We use greedy decoding with temperature = 0.0, `max_new_tokens`= 64 and thinking_mode disabled for all models.

A.2.1 Paired Inputs Definition: In EMOTIC, each annotated subject has its own bounding-box/marker and emotion labels. If one image contains several annotated subjects, D2 evaluates each annotated subject separately. In CAER, the emotion label is attached to the main subject of the whole clip, so D2 evaluates each clip once. For each annotated subject or clip i, we use the same prompt and compare

$$V_i \quad \text{vs.} \quad \widetilde{V}_i^{\,b}. \tag{3}$$

Here, b is the target-subject box in EMOTIC and the selected pseudo person box in CAER. For CAER, both inputs use the same four sampled frames, and the box is applied to the selected track in each frame. A pair is kept only when both model outputs are valid dataset labels.

A.2.2 Prediction-Change and Distributional Summaries: For each valid pair, we parse one single-label prediction from the original visual input and one from the subject-removed input:

$$\begin{aligned} y_i^*(V_i) &= \arg\max_{y\in\mathcal{Y}} p_{\mathcal{M}}(y \mid V_i, T), \\ y_i^*(\widetilde{V}_i^{\,b}) &= \arg\max_{y\in\mathcal{Y}} p_{\mathcal{M}}(y \mid \widetilde{V}_i^{\,b}, T). \end{aligned} \tag{4}$$

A prediction is counted as correct when the parsed label matches the official annotation(s) of the same annotated subject or clip.

The prediction-change rate is computed on the same valid pairs:

$$\frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left[\arg\max_{y\in\mathcal{Y}} p_{\mathcal{M}}(y \mid V_i, T) \neq \arg\max_{y\in\mathcal{Y}} p_{\mathcal{M}}(y \mid \widetilde{V}_i^{\,b}, T)\right]. \tag{5}$$

For each of V_i and $\widetilde{V}_i^{\,b}$, we also summarize the empirical distribution of parsed predictions over the same valid pairs. We report the most frequent label and its proportion, the

proportions of the three most frequently predicted labels, and the entropy-effective number N_{eff}, computed as the exponential of the empirical prediction entropy.

A.2.3 Control Contrasts Setting for Diagnosis 2: First, we test the effect of adding a gray rectangle outside the subject region. For each sample, we enumerate all candidate top-left ranges that place a rectangle with the same width and height as b_i completely to the left, to the right, above, or below b_i, while keeping it inside the image. If a candidate range contains n_q integer top-left positions, that range is sampled with probability

$$\frac{n_q}{\sum_{q'} n_{q'}}. \tag{6}$$

The top-left position is then sampled uniformly inside the selected range using a fixed seed derived from the sample identifier. Let $\bar{b}_i$ denote the selected feasible region:

$$\bar{b}_i \cap b_i = \emptyset. \tag{7}$$

If no such region exists, the sample is not used for this control. We construct $V_i^{\bar{b}_i}$ from V_i by keeping all pixels in the subject box visible and replacing only the pixels inside $\bar{b}_i$ with the same gray value used in $\widetilde{V}_i^b$. The contrast V_i versus $V_i^{\bar{b}_i}$ measures the effect of adding the gray rectangle to the original visual input.

Second, we test whether the subject-removal effect comes from covering the subject region rather than from the gray rectangle itself. We compare $\widetilde{V}_i^b$ with $V_i^{\bar{b}_i}$. Both inputs contain one gray rectangle of the same size and value; the difference is whether it covers the subject box or the non-overlapping region $\bar{b}_i$.

Third, we remove the explicit visual marker from the original visual input on eligible EMOTIC samples:

$$V_i \quad \text{vs.} \quad V_i^{\text{mf}}. \tag{8}$$

The superscript "mf" denotes the marker-free version of the same input.

Fourth, we remove the explicit visual marker from the subject-removed input on the same eligible EMOTIC samples:

$$\widetilde{V}_i^b \quad \text{vs.} \quad (\widetilde{V}_i^b)^{\text{mf}}. \tag{9}$$

Moreover, for EMOTIC, we stratify the same V_i versus $\widetilde{V}_i^b$ comparison by the number of annotated subjects in the image: one, two, and at least three. This tests whether subject removal has the same effect when other annotated subjects remain visible.

Besides, for CAER, we repeat subject removal with the localized pseudo face/head box described in SM G. The comparison is still V_i versus $\widetilde{V}_i^b$; only the region instantiated by b is changed.

A.3 Detailed Setting for Diagnosis 3 (Causal Mirage II)

A.3.1 Cross-Sample Pairing Rule: D3 first pairs each target sample i with a sample j that carries different visual evidence. To ensure that the two samples provide distinguishable affective evidence, we pair each sample i with a sample j having a sufficiently different emotion.

First, since EMOTIC provides valence, arousal, and dominance (VAD) ratings, we build the rule based on VAD. j must come from a different image and share no emotion annotation with sample i:

$$\text{img}(j) \neq \text{img}(i). \tag{10}$$

Within this eligible set, we define the VAD distance of a candidate pair using the EMOTIC valence, arousal, and dominance ratings. Let $\mathbf{z}_i$ be the VAD rating vector of sample i:

$$\mathbf{z}_i = (z_i^{\text{val}}, z_i^{\text{aro}}, z_i^{\text{dom}}). \tag{11}$$

Choosing the farthest candidate independently for each i would repeatedly use a small number of samples with large VAD distance from many targets. **In EMOTIC, this would use only 45 distinct paired samples and reuse one paired sample 4439 times.** We therefore choose the set of pairs together. For every target sample i, the pairing rule selects one j while constraining each paired sample to be used at most seven times. Under this constraint, the selected pairs maximize the sum of normalized VAD distances:

$$\sum_i \left\| \frac{\mathbf{z}_i - \mathbf{z}_j}{9} \right\|_2, \tag{12}$$

where the division by 9 normalizes the rating range. The limit of seven is the smallest value that still gives every EMOTIC target-person annotation a valid paired sample under the constraints above. The resulting pairs cover all 10,613 subject annotations and use 1,583 distinct paired samples.

Second, since CAER utilizes Ekman's [1] 7 basic emotions $\mathcal{Y}_{\text{CAER}} = \{\text{neutral, happy, sad, anger, surprise, disgust, fear}\}$, where we can not find an opposite emotion for each label, we define the rule based on semantics. Following the formulation of PAD [2], in which pleasure, arousal, and dominance (PAD) are together necessary to characterize an emotional state, we form pairs that are each opposed along one factor: *happy–sad* along pleasure, *neutral–surprise* along arousal, and *anger–fear* along dominance. Because $|\mathcal{Y}_{\text{CAER}}|$ is odd, *disgust* has no remaining counterpart and is paired with *happy* along pleasure. Note that we do not assign every category to its most dissimilar emotion in PAD space; otherwise, every category would be paired with either *happy* or *sad*. Given this category-level rule, the remaining step is to form clip-level pairs within each split. Each target clip i is paired with one clip j from the same split and from its paired category in the rule above. Within each target category, clips are ordered by sample identifier and paired in a round-robin order with clips from the paired category. This covers all 3,953 CAER clips, uses 2,340 distinct paired clips, and uses no paired clip more than four times. The same paired clip j is used to construct both crossed visual inputs.

[1] Ekman, P. (1992). Facial Expressions of Emotion: New Findings, New Questions. Psychological Science, 3(1), 34-38.

[2] Russell, J. A., & Mehrabian, A. (1977). Evidence for a three-factor theory of emotions. Journal of research in Personality, 11(3), 273-294.

A.3.2 Evaluation Anchor and Paired Records: For each target sample i, j denotes the paired sample selected by the pairing rule above. All D3 recognition scores are evaluated against the official annotation(s) of i. For each pair, we evaluate

$$V_i = (S_i, C_i), \qquad (S_i, C_j), \qquad (S_j, C_i). \tag{13}$$

The two D3 comparisons are V_i versus (S_i, C_j), which changes the scene context, and V_i versus (S_j, C_i), which changes the subject region. A sample is kept for a comparison only when both model outputs are valid dataset labels. We write $y_i^*(\cdot)$ for the parsed prediction under the visual input in parentheses. The prediction-change rates are

$$\begin{aligned} (S_i, C_j): &\quad \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}[y_i^*(V_i) \neq y_i^*(S_i, C_j)], \\ (S_j, C_i): &\quad \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}[y_i^*(V_i) \neq y_i^*(S_j, C_i)], \end{aligned} \tag{14}$$

where N is the number of retained target samples in the corresponding comparison. Recognition changes use the same retained samples and are reported as crossed-input accuracy minus V_i accuracy.

A.3.3 Control Contrasts Setting for D3: The D3 controls keep the same pairing rule and the same evaluation anchor, so all scores are still measured against sample i. First, on single-person EMOTIC samples, we repeat subject-region replacement with and without the explicit visual marker:

$$(S_j, C_i) \quad \text{with and without the marker.} \tag{15}$$

Second, on CAER, we repeat the two crossed inputs with the localized pseudo face/head region instead of the pseudo person region. Let S_i^{fh} denote this localized region and C_i^{fh} denote the remaining visual content. The controlled crossed inputs are

$$(S_i^{\text{fh}}, C_j^{\text{fh}}), \qquad (S_j^{\text{fh}}, C_i^{\text{fh}}). \tag{16}$$

A.4 Detailed Setting for Diagnosis 4 (Certainty Mirage)

D4 uses FER+ over an eight-class taxonomy (the seven canonical classes plus *contempt*), where each image carries votes from ten annotators over the emotion categories. Images whose kept-class votes sum below five, or whose argmax is an *unknown/not-a-face* column, are dropped. The per-image human distribution $\mathbf{h}_i$ is the normalized vote vector, and human uncertainty is its Shannon entropy (base 2, theoretical):

$$H(\mathbf{h}_i) \;=\; -\sum_{c} h_{i,c}\,\log_2 h_{i,c}. \tag{17}$$

Human agreement is summarized by the top vote share $h_i^{\text{human}} = \max_c h_{i,c}$, used to form agreement groups (below).

Model confidence via teacher forcing. Because the open models expose token logits, we obtain a model distribution over the eight classes by teacher-forced scoring rather than sampling. For class c with a set of surface forms $\mathcal{V}_c$ (verbalizers; only *anger* realizes two forms, `angry`/`anger`), the class score marginalizes over surface forms in log space and is normalized by a softmax:

$$s_{i,c} \;=\; \log\sum_{v\in\mathcal{V}_c} \exp \ell_i(v), \qquad p_{i,c} \;=\; \frac{\exp s_{i,c}}{\sum_{c'} \exp s_{i,c'}}, \tag{18}$$

where $\ell_i(v)$ is the model log-likelihood of surface form v under the D4 prompt (Appendix G.5). Model confidence is $p_i^{\text{top1}} = \max_c p_{i,c}$ and model uncertainty is the entropy $H(\mathbf{p}_i)$ as in Eq. (17). Closed-source models lack accessible output heads, so their D4 numbers are reported for reference only and excluded from comparisons.

Alignment between model and human uncertainty is measured by the Spearman rank correlation ρ between the model entropy $H(\mathbf{p}_i)$ and the human entropy $H(\mathbf{h}_i)$, matching the main-text definition. We report three variants: ρ_{all} over all samples; ρ_{correct} over the correctly-predicted subset; and a first-order *partial* Spearman controlling for prediction correctness z_i,

$$\rho_{H_M H_H|z} \;=\; \frac{\rho_{H_M H_H} - \rho_{H_M z}\,\rho_{H_H z}}{\sqrt{(1-\rho_{H_M z}^2)(1-\rho_{H_H z}^2)}}, \tag{19}$$

which isolates alignment not explained by prediction correctness. Each ρ is accompanied by a bootstrap 95% confidence interval. The model confidence p_i^{top1} and the human agreement h_i^{human} are used only for the high/low agreement analysis below.

Following the main text, samples are split into a **high**-agreement group H ($h^{\text{human}} \geq 0.8$, the median 0.800) and a **low**-agreement group L ($h^{\text{human}} \leq 0.5$, samples without a clear majority). For each group we report accuracy and p^{top1} (Table IV); a calibrated model should show lower confidence in L.

To calibrate the achievable ρ, we estimate a split-half reliability: the votes of each image are split into two halves by a multivariate-hypergeometric draw, and ρ between the two half-entropies is averaged over 100 splits on the FER+ evaluation set (3,546 images, base-2 entropy), giving $\rho_{\text{half}} = 0.510$ (95% CI [0.495, 0.523]). Because each half holds only about five votes, we apply the Spearman–Brown correction to obtain the reliability of the full ten-vote entropy,

$$\rho_{\text{ceiling}} \;=\; \frac{2\rho_{\text{half}}}{1+\rho_{\text{half}}} \;=\; 0.675 \quad (95\%\ \text{CI}\ [0.662, 0.687]). \tag{20}$$

Against this ceiling, Qwen3.5-9B ($\rho_{\text{all}} = 0.601$; Table I) and InternVL3.5-8B (0.548) reach 89% and 81%, Emotion-LLaMA 54%, and Ministral-3-8B 34%.

Calibration and temperature scaling. Correctness is agreement of the arg-max class of $\mathbf{q}_i$ with the human majority label, so prediction and confidence come from the same distribution. ECE uses 10 equal-width bins of p^{top1}; the Brier score is $\sum_c (q_{i,c} - \mathbb{1}[c = y_i])^2$ averaged over images; JSD is computed between $\mathbf{q}_i$ and $\mathbf{h}_i$ (base 2). For temperature scaling, $q_{i,c}(T) \propto \exp(\log q_{i,c}/T)$, with one T per model fitted by minimizing the negative log-likelihood of the majority label on four folds and applied to the held-out fold (5-fold cross-validation); the arg-max, and hence

accuracy, is unchanged. Table S1 lists T and the metrics before and after scaling. Scaling removes most of the global miscalibration (ECE 0.02–0.09), yet for the four positively aligned models the low-agreement gap stays at $+0.17$ to $+0.21$ while the high-agreement group turns underconfident. No single temperature fits both groups: the models do not lower their confidence specifically where humans disagree.

B. Further Discussion

B.1 Discussion: Why AffectGPT Behaves as a Negative Control

Across all four diagnoses, AffectGPT departs sharply from the other six models, and we treat it as a model-specific fitting issue that does not affect the overall trend. This section explains the behavior and shows it is a *clean negative control* rather than a counterexample to MIRAGE.

We evaluated AffectGPT under two interfaces. (i) A native single-image route yields degenerate generations: on RAF-DB it reaches an accuracy of only 0.182 under the image-present condition (Table I) and produces malformed outputs on the majority of no-vision runs, surfacing as a high parsing-failure share in Table S5. (ii) Same interface across all diagnoses: present accuracy 0.182 and blank accuracy 0.210 are nearly identical ($\Delta \approx +0.03$), and the present-vs-blank prediction distributions are almost indistinguishable (Jensen–Shannon divergence ≈ 0.005). Under either route the face carries essentially no information about the emotion decision, so AffectGPT satisfies the definition of a negative control.

The inert-vision reading is corroborated downstream. In D1 the multi-seed mirage-hint condition almost never induces a change ($1/50$ seeds), i.e., the model neither grounds on the face nor reacts to the presence cue. In D4 its model-vs-human entropy correlation is *negative* ($\rho_{\text{all}} = -0.245$) and stays near zero on the correct subset ($\rho_{\text{correct}} = -0.077$), the opposite of the four positively-aligned open models. Its confidence is also pathologically high ($p^{\text{top1}} = 0.992$, the largest of all models), and its accuracy ordering across agreement groups is *inverted* (Table IV: Acc.$H = 0.329 <$ Acc.$L = 0.393$): it is no more accurate on high-agreement (easy) images than on low-agreement (hard) ones, which is precisely what a model that ignores visual evidence and emits a near-constant high-confidence prediction would produce.

Both access routes converge on the same conclusion: the visual pathway does not drive AffectGPT's emotion output. So its numbers should be read as a floor / negative control that anchors the MIRAGE scale, not as evidence against the phenomenon. Its non-neutral collapse pattern (parsing failures rather than a clean neutral prior) is discussed alongside Emotion-LLaMA in the main text.

B.2 Discussion of prediction entropy (D4 model vs human uncertainty)

Beyond the rank correlations of Table I, we examine *where* on the uncertainty scale the model and human distributions agree, using the per-sample entropy of Eq. (17) (Fig. 4).

General models underestimate human affective ambiguity. For the three general VLMs, many samples with high human entropy fall *below* the diagonal: model uncertainty stays low even where annotators strongly disagree. This is consistent with Table IV, where these three general VLMs retain high confidence ($p_L^{\text{top1}} = 0.72$–0.79) while their accuracy collapses on the low-agreement group (Acc.$L = 0.36$–0.47 vs. Acc.$H = 0.74$–0.94). The models capture a coarse ranking of sample difficulty but systematically compress the *level* of uncertainty relative to humans (rank alignment without matched confidence).

Emotion-LLaMA exhibits broadly higher entropy, including on low-human-entropy (easy) samples, and has the lowest overall confidence ($p^{\text{top1}} = 0.547$); it spreads probability mass rather than committing. AffectGPT does the opposite, remaining concentrated at very low model entropy ($p^{\text{top1}} = 0.992$) regardless of human agreement, consistent with its negative-control reading in B.1.

The high-agreement threshold $h^{\text{human}} \geq 0.8$ coincides with the sample median and is close to the theoretical concentration anchor $1 - 1/K = 0.875$ for $K = 8$ classes (a distribution whose top label carries essentially all the mass). The low-agreement threshold $h^{\text{human}} \leq 0.5$ identifies samples *without a clear majority*; it cannot be pushed toward the uniform value $1/K = 0.125$ because too few samples are that ambiguous, so 0.5 is the data-forced cut for a populated low bin. These thresholds define the H/L groups used in Table IV.

Model uncertainty increases with human disagreement (positive ρ), but confidence does not decrease enough on low-agreement images (Table IV), even after temperature scaling (A.4), so alignment with human affective ambiguity is only partial — the certainty-mirage counterpart of the label-prior collapse seen in D1.

C. Additional Results for General Mirage

A natural concern is that requiring the model to choose from a fixed list of options forces it to emit an emotion label, so a prediction produced without visual evidence need not reflect a genuine visual hallucination. We therefore test whether the general mirage of D1 survives four prompt design changes, each of which removes part of the option-menu constraint or explicitly allows abstention. Throughout we report the same quantities as the main text: the four mutually exclusive outcome rates (collapse to *neutral*, off-neutral emotion, parsing failure, abstention), the Mirage Rate $R_{\text{mirage}} = 1 -$ abstention, and the Collapse Rate R_{collapse}, with the 0.222 RAF-DB neutral base rate as the no-vision accuracy anchor. Single-seed and multi-seed values are shown as seed 0 / mean over S=50 seeds.

C.1 Open-ended prompt

We remove the option menu and offer no abstention option, asking only which emotion the person is expressing and requesting the answer in the frozen bracket format (Appendix G.2). A named emotion counts as a mirage, whereas a spontaneous statement that no person or image is present counts as an abstention. Non-committal tokens such as *unclear* or *unknown* are counted conservatively as

TABLE S1

D4: 5-FOLD CROSS-VALIDATED TEMPERATURE SCALING (FER+, OPEN-SOURCE MODELS). ENTRIES ARE BEFORE → AFTER SCALING; GAP $= \bar{p}_{\text{top1}} - \text{Acc}$ (> 0: OVERCONFIDENT) WITHIN THE HIGH (H) AND LOW (L) AGREEMENT GROUPS. T IS THE MEAN OVER FOLDS.

Model	T	ECE	Brier	JSD	ρ_{all}	gap H	gap L
Qwen3.5-9B	1.29	0.057→0.019	0.297→0.289	0.130→0.126	0.601→0.614	−0.013→−0.051	+0.251→+0.181
InternVL3.5-8B	2.07	0.132→0.020	0.431→0.400	0.202→0.193	0.548→0.535	+0.028→−0.096	+0.369→+0.178
Ministral-3-8B	2.48	0.215→0.058	0.619→0.559	0.290→0.295	0.233→0.310	+0.112→−0.120	+0.429→+0.170
Emotion-LLaMA	1.11	0.106→0.092	0.629→0.624	0.364→0.367	0.365→0.367	+0.011→−0.023	+0.236→+0.205
AffectGPT	5.84	0.633→0.061	1.263→0.778	0.592→0.474	−0.245→0.111	+0.660→+0.089	+0.603→+0.028

TABLE S2

C.1: OPEN-ENDED PROMPT (NO MENU, NO ABSTENTION OPTION). OUTCOME COMPOSITION AND MIRAGE RATE UNDER BLANK AND NOISE, SEED 0 / MEAN OVER S=50. ONLY *Abst* IS NON-MIRAGE.

Model	Cond.	Col. to neu	Off-neu	P-fail	Abst	R_{mirage}
GPT-5.6-Luna	Blank	1/1	0/0	0/0	0/0	1/1
	Noise	0.34/0.27	0.66/0.73	0/0	0/0.01	1/0.99
Gemini-3.5-Flash-Lite	Blank	1/1	0/0	0/0	0/0	1/1
	Noise	1/0.88	0/0.12	0/0	0/0	1/1
Qwen3.5-9B	Blank	1/0.58	0/0.42	0/0	0/0	1/1
	Noise	0/0.38	1/0.60	0/0	0/0.02	1/0.98
InternVL3.5-8B	Blank	0/0.36	1/0.64	0/0	0/0	1/1
	Noise	0/0.14	1/0.86	0/0	0/0	1/1
Ministral-3-8B	Blank	1/0.62	0/0.14	0/0.10	0/0.14	1/0.86
	Noise	0/0.02	1/0.75	0/0.12	0/0.11	1/0.89
Emotion-LLaMA	Blank	1/0.38	0/0.38	0/0.18	0/0.06	1/0.94
	Noise	0/0.16	0/0.54	1/0.24	0/0.06	1/0.94
AffectGPT	Blank	0/0.08	1/0.69	0/0.23	0/0	1/1
	Noise	0/0.01	1/0.73	0/0.26	0/0	1/1

mirage, since they do not explicitly recognize that the image is absent; only explicit statements of missing evidence count as abstention.

As shown in Table S2, across all seven models the abstention rate stays at or near zero and R_{mirage} stays at or near one, under both blank and noise and for both the single seed and the S=50 mean. Removing the option menu does not recover evidence-aware behavior: the models continue to assert an emotion instead of reporting that the image is absent. The collapse target is model specific, with some models falling back to *neutral* and others to a non-neutral emotion, which is consistent with the label-prior collapse of D1.

C.2 Unrelated natural image

We replace the visibly degenerate blank and noise inputs with a fixed natural image that contains no person, keeping the basic prompt (G.2). Table S3 reports the collapse-to-neutral rate and the Mirage Rate.

Every model collapses entirely to *neutral* (collapse rate 1), so its no-vision accuracy equals the neutral base rate of 0.222, and R_{mirage} remains 1. The mirage is thus not an artifact of visibly degenerate placeholders: an ordinary image with no face elicits the same neutral prior as a blank or noise input.

TABLE S3

C.2: UNRELATED NATURAL IMAGE (BASIC PROMPT, SEED 0). EVERY MODEL COLLAPSES TO *neutral*, SO NO-VISION ACCURACY EQUALS THE 0.222 NEUTRAL BASE RATE.

Model	Col. to neu	R_{mirage}
GPT-5.6-Luna	1	1
Gemini-3.5-Flash-Lite	1	1
Qwen3.5-9B	1	1
InternVL3.5-8B	1	1
Ministral-3-8B	1	1
Emotion-LLaMA	1	1
AffectGPT	1	1

TABLE S4

C.3: LABEL-ORDER PERMUTATION (BLANK, S=50). R_{COLLAPSE} TOWARD *neutral* PER ORDERING AND ITS MEAN±STD OVER FIVE PERMUTATIONS.

Model	ord. 0	ord. 1	ord. 2	ord. 3	ord. 4	mean±std
Qwen3.5-9B	0.96	0.96	0.96	0.96	0.96	0.96±0.00
InternVL3.5-8B	0.90	1	0.92	0.90	0.94	0.93±0.04
Ministral-3-8B	0.90	0.71	0.80	0.92	0.82	0.83±0.08
Emotion-LLaMA	0.60	0.86	0.54	0.38	0.36	0.55±0.18
AffectGPT	0.48	0.43	0.50	0.05	0.34	0.36±0.18

C.3 Label-order permutation

We randomize the order of the seven options across five fixed permutations that place *neutral* in different positions, keeping the blank input and the basic prompt. Table S4 reports R_{collapse} toward *neutral* per ordering.

For the general models the collapse rate is stable across permutations (0.96, 0.93, and 0.83 on average for Qwen3.5-9B, InternVL3.5-8B, and Ministral-3-8B, with small dispersion), so the collapse target is not induced by the position of the options. The emotion-specialized models show a lower and more variable collapse rate, because their frequent parsing failures scatter the recovered label; their R_{mirage} nonetheless stays high across all orderings.

C.4 Abstention option

We keep the basic prompt and only append *cannot determine* as an eighth option (AO prompt, G.2). A response counts as an abstention only if the selected option is *cannot determine*; every other output, including parsing failures, is a mirage. AO is run on the three no-vision inputs (S=50 seeds for open models, 50 independent calls for closed-source models, whose third input sends no image) and on

the full test set for *image_present*, where any abstention is a false abstention. Table S5 reports the outcome composition and Table S6 the real-face results; the basic-prompt rows are the multi-seed counterparts of Table I.

Both closed-source models abstain on every no-vision call while abstaining on only 2–4% of real faces and keeping their accuracy, so they can detect the missing evidence but do not report it proactively under the basic prompt. The open-source models do not abstain in an evidence-aware way. Qwen3.5-9B still answers on 14–40% of no-vision runs. InternVL3.5-8B abstains on 98% of noise runs but answers on 86% of blank runs, mostly with *neutral*: it treats a uniform gray image as a neutral face rather than as missing evidence. Ministral-3-8B abstains on almost all no-vision runs but also on 43.7% of real faces, which lowers its accuracy from 0.546 to 0.317 while its answered-only accuracy stays at 0.562; for InternVL3.5-8B the false-abstention rate is 12.9%. Emotion-LLaMA never selects the option and keeps its basic-prompt behavior, including 32–58% *neutral* and the remaining mass on other emotions. AffectGPT likewise never selects the option: its no-vision outputs concentrate further on *neutral* (80–86%, with 12–18% parsing failures), and on real faces it predicts *neutral* for 99.5% of images, consistent with its negative-control reading (SM B.1).

C.5 Summary

After the models are queried with an open-ended prompt (C.1), shown an ordinary image with no face (C.2), and tested across option orderings (C.3), they still almost never abstain and still fall back to a label prior. When abstention is explicitly offered (C.4), both closed-source models abstain on every no-vision input while rarely abstaining on real faces, so they can detect missing evidence but do not report it proactively under the basic prompt. Open-source and emotion-specific models, however, keep answering on blank inputs, abstain on real faces, or never abstain. The near-100% mirage rate reported in D1 therefore reflects a genuine failure to proactively signal the absence of visual evidence, and for open-source models even an explicit option does not make abstention evidence-aware.

D. Additional Recognition Results for Causal Mirage

This section expands the D2 and D3 recognition results reported in the main paper using both validation and test splits. Following Sec. 3, $V_i = (S_i, C_i)$ denotes the original input, $\widetilde{V}_i^b$ the subject-removed input, (S_i, C_j) scene-context replacement, and (S_j, C_i) subject-region replacement. The additional analyses examine whether residual recognition after subject removal reflects scene-context substitution (D2) and which visual evidence dominates under subject-scene mismatch (D3).

D.1 Diagnosis 2: Causal Mirage in Subject Removal

D2 expands the subject-removal result by jointly examining recognition performance and prediction distributions. Table S7 reports prediction-distribution summaries for all D2 and D3 visual inputs, with annotation-frequency results added as baselines. These baseline rows are not model outputs. For EMOTIC, they report the fraction of annotated subjects whose ground-truth label set contains each label. For CAER, they report the fraction of clips assigned to each label. They give the dominant-label scale for reading prediction concentration.

N_{eff} converts the prediction entropy into the effective number of labels used by the model. Larger values mean that predictions are spread over more labels. The Distinct labels row is a looser count: it records how many labels appear at least once, so it can remain high even when most predictions fall on a small subset of labels.

Table S8 reports how often the parsed prediction changes relative to V_i. Under D2, subject removal changes 41.6%–74.9% of predictions for the five general VLMs on EMOTIC and 45.9%–80.4% on CAER. This shows that the intervention changes the output label for many samples; the recognition and distribution tables then show whether the changed predictions remain correct and how concentrated they become

Table S7 adds two distributional details to the main-text D2 result. First, **the five general VLMs show related concentration patterns after subject removal**. On EMOTIC, N_{eff} decreases for all five models and the Top-3 share increases, while the number of distinct predicted labels changes only mildly. Low-frequency labels therefore remain present, but more prediction mass moves into a smaller set of preferred labels. On CAER, the same concentration is much sharper: after subject removal, the five general VLMs have N_{eff} between 1.1 and 1.7, and their Top-3 shares range from 98.2% to 99.5%. This is far more concentrated than the CAER annotation baseline, where the most frequent label accounts for 34.7% of clips.

Second, **the two affective VLMs do not follow the shared pattern of the general VLMs**. AffectGPT is already highly concentrated under V_i, so its small change after subject removal reflects a concentrated starting point rather than reliable use of the remaining visual evidence. Emotion-LLaMA also uses a small effective label set, but its concentrated labels differ from those of the general VLMs, especially on CAER. These results therefore indicate model-specific collapse patterns for the affective VLMs, rather than a shared pattern across this model group.

The Macro-F1 rows in Table S9 add the class-balanced view that is only summarized in the main text. Because Macro-F1 gives each label equal weight, it drops when a model preserves part of its Hit@1/Acc@1 through frequent labels but loses coverage over the label set. For the five general VLMs, EMOTIC Macro-F1 falls from 0.128–0.148 under V_i to 0.060–0.091 under $\widetilde{V}_i^b$. On CAER, it falls from 0.275–0.379 to 0.086–0.116. These Macro-F1 results therefore support the distribution summaries: residual Hit@1/Acc@1 after subject removal is not broad recognition across emotion labels.

TABLE S5

C.4: OUTCOME COMPOSITION WITHOUT VISION, BASIC VS. AO PROMPT. EACH ENTRY IS SEED 0 / MEAN OVER S=50 SEEDS (CLOSED-SOURCE MODELS: FIRST CALL / MEAN OVER 50 CALLS; *No-img* SENDS NO IMAGE). *Col. to neu*: *neutral* OUTPUT; *Off-neu*: OTHER OUTPUT, INCLUDING OFF-TAXONOMY TOKENS; *P-fail*: PARSING FAILURE; *Abst*: ABSTENTION. ONLY *Abst* IS NON-MIRAGE.

Model	Input	Prompt	Col. to neu	Off-neu	P-fail	Abst	R_{collapse}	R_{mirage}
GPT-5.6-Luna	Blank	Basic	1/1	0/0	0/0	0/0	1/1	1/1
		AO	0/0	0/0	0/0	1/1	0/0	0/0
	Noise	Basic	1/1	0/0	0/0	0/0	1/1	1/1
		AO	0/0	0/0	0/0	1/1	0/0	0/0
	No-img	Basic	1/1	0/0	0/0	0/0	1/1	1/1
		AO	0/0	0/0	0/0	1/1	0/0	0/0
Gemini-3.5-Flash-Lite	Blank	Basic	1/1	0/0	0/0	0/0	1/1	1/1
		AO	0/0	0/0	0/0	1/1	0/0	0/0
	Noise	Basic	1/1	0/0	0/0	0/0	1/1	1/1
		AO	0/0	0/0	0/0	1/1	0/0	0/0
	No-img	Basic	1/1	0/0	0/0	0/0	1/1	1/1
		AO	0/0	0/0	0/0	1/1	0/0	0/0
Qwen3.5-9B	Blank	Basic	1/0.96	0/0	0/0.02	0/0.02	1/0.96	1/0.98
		AO	0/0.28	0/0	0/0	1/0.72	0/0.28	0/0.28
	Noise	Basic	1/0.98	0/0	0/0	0/0.02	1/0.98	1/0.98
		AO	0/0.14	0/0	0/0	1/0.86	0/0.14	0/0.14
	Hint	Basic	1/0.88	0/0	0/0.04	0/0.08	1/0.88	1/0.92
		AO	0/0.38	0/0	0/0.02	1/0.60	0/0.38	0/0.40
InternVL3.5-8B	Blank	Basic	1/0.90	0/0.10	0/0	0/0	1/0.90	1/1
		AO	0/0.78	0/0.08	0/0	1/0.14	0/0.78	0/0.86
	Noise	Basic	1/0.96	0/0.04	0/0	0/0	1/0.96	1/1
		AO	0/0.02	0/0	0/0	1/0.98	0/0.02	0/0.02
	Hint	Basic	1/0.88	0/0.12	0/0	0/0	1/0.88	1/1
		AO	0/0.70	0/0.08	0/0	1/0.22	0/0.70	0/0.78
Ministral-3-8B	Blank	Basic	1/0.90	0/0.02	0/0.02	0/0.06	1/0.90	1/0.94
		AO	0/0.04	0/0	0/0	1/0.96	0/0.04	0/0.04
	Noise	Basic	1/0.86	0/0.01	0/0.01	0/0.12	1/0.86	1/0.88
		AO	0/0	0/0	0/0	1/1	0/0	0/0
	Hint	Basic	1/0.94	0/0	0/0	0/0.06	1/0.94	1/0.94
		AO	1/0.22	0/0	0/0	0/0.78	1/0.22	1/0.22
Emotion-LLaMA	Blank	Basic	1/0.60	0/0.40	0/0	0/0	1/0.60	1/1
		AO	1/0.58	0/0.42	0/0	0/0	1/0.58	1/1
	Noise	Basic	1/0.38	0/0.60	0/0.02	0/0	1/0.38	1/1
		AO	1/0.32	0/0.68	0/0	0/0	1/0.32	1/1
	Hint	Basic	1/0.60	0/0.40	0/0	0/0	1/0.60	1/1
		AO	1/0.46	0/0.54	0/0	0/0	1/0.46	1/1
AffectGPT	Blank	Basic	0/0.44	0/0	1/0.54	0/0.02	0/0.44	1/0.98
		AO	1/0.80	0/0.02	0/0.18	0/0	1/0.80	1/1
	Noise	Basic	1/0.54	0/0	0/0.46	0/0	1/0.54	1/1
		AO	1/0.86	0/0.02	0/0.12	0/0	1/0.86	1/1
	Hint	Basic	0/0.30	0/0.02	1/0.68	0/0	0/0.30	1/1
		AO	1/0.84	0/0.02	0/0.14	0/0	1/0.84	1/1

D.2 Diagnosis 3: Causal Mirage in Subject–Scene Mismatch

D3 expands the main-text subject evidence dominance result by reading the crossed visual inputs together with the prediction distributions. The additional question is whether the D3 effects reproduce the strong concentration observed after subject removal, or whether the crossed inputs keep a broader prediction distribution while subject-region evidence and scene context disagree.

For D3, the relevant results are (S_i, C_j) and (S_j, C_i) in Tables S7 and S9. The distribution results show that the crossed inputs retain substantial prediction diversity when subject-region evidence and scene context are mismatched. On CAER, the five general VLMs have N_eff between 3.6 and 5.2 under the two crossed inputs and retain six or seven distinct predicted labels. On EMOTIC, subject-region replacement often makes the predictions more dispersed. Thus, the crossed interventions preserve nontrivial variation in model predictions despite the disagreement between the two evidence sources.

At the same time, subject-region replacement is substantially more disruptive than scene-context replacement. Across the five general VLMs, (S_i, C_j) keeps Macro-F1 much closer to V_i, whereas under (S_j, C_i) it falls to 0.031–0.057 on EMOTIC and 0.068–0.091 on CAER. **The coexistence of substantial prediction diversity and markedly larger degradation under subject-region replacement provides recognition-level evidence for subject-region dominance.**

D.3 Summary

Overall, D2 shows that residual recognition after subject removal should not be interpreted as effective scene-context substitution, as it is accompanied by prediction concentration and reduced Macro-F1. D3 instead shows broader prediction

TABLE S6

C.4: REAL FACES (*image_present*, FULL RAF-DB TEST SET), BASIC VS. AO PROMPT. F-ABST: FALSE ABSTENTION; ACC (ANS.): ACCURACY OVER ANSWERED IMAGES ONLY. 8 REAL-FACE IMAGES REFUSED BY THE PROVIDERS' CONTENT FILTERS ARE EXCLUDED (GPT-5.6-LUNA 7, GEMINI-3.5-FLASH-LITE 1).

Model	Prompt	Acc	Macro-F1	F-abst	Acc (ans.)
GPT-5.6-Luna	Basic	0.737	0.610	0.0%	0.737
	AO	0.738	0.636	2.3%	0.755
Gemini-3.5-Flash-Lite	Basic	0.646	0.546	0.0%	0.646
	AO	0.661	0.574	3.9%	0.687
Qwen3.5-9B	Basic	0.692	0.554	0.0%	0.692
	AO	0.708	0.584	0.2%	0.710
InternVL3.5-8B	Basic	0.762	0.637	0.0%	0.762
	AO	0.685	0.617	12.9%	0.786
Ministral-3-8B	Basic	0.546	0.426	0.0%	0.546
	AO	0.317	0.344	43.7%	0.562
Emotion-LLaMA	Basic	0.747	0.513	0.0%	0.747
	AO	0.745	0.512	0.0%	0.745
AffectGPT	Basic	0.182	0.053	4.0%	0.190
	AO	0.227	0.056	0.0%	0.227

distributions but stronger degradation under subject-region replacement. Together these results support the conclusion that scene context rarely substitutes for person-specific affective evidence under subject-scene mismatch.

E. CONTROLLED ANALYSES FOR CAUSAL MIRAGE

This section reports controls that support the D2 and D3 interpretations in the main text. The D2 controls test whether subject removal is driven by the gray rectangle, the explicit visual marker, other annotated subjects in the image, or the size of CAER pseudo subject-boxes. The D3 controls test whether subject-region replacement remains stronger than scene-context replacement after removing the marker or localizing the CAER subject region to the face/head.

E.1 Diagnosis 2: Subject-Removal Controls

First, for D2, the gray-rectangle controls in Table S10 separate the effect of the rectangle from the effect of masking subject-region visual evidence. The number of samples in this control subset (i.e., the equal-area feasible subset) is smaller than the full dataset because only samples with sufficient non-subject image area can accommodate an equal-sized gray rectangle. Under this rating, adding the same gray rectangle away from the subject to V_i changes Hit@1 by -0.6 to $+0.0$ points for Qwen3.5-9B, InternVL3.5-8B, and Ministral 3 8B. In contrast, keeping the subject visible and placing the gray rectangle away from the subject gives $+24.4$ to $+41.4$ points relative to $\widetilde{V}_i^b$. Thus the prediction collapse is not caused by the equal-area gray rectangle alone.

Second, the D2 marker results show a smaller effect. Removing the explicit visual marker from V_i changes Hit@1 by $+0.4$, $+0.3$, and $+0.0$ points for the same three models. Removing it from $\widetilde{V}_i^b$ changes Hit@1 by $+7.1$, $+7.5$, and $+1.7$ points. The marker can affect the subject-removed input, but these changes are much smaller than the subject-removal changes in Table S9, suggesting that the marker is not the main reason for prediction collapse.

Besides, Table S11 stratifies D2 by the number of annotated subjects in the image. The three groups have different sample sizes because they follow the natural EMOTIC distribution: around 3900 samples contain one annotated subject, around 3350 contain two annotated subjects, and around 3357 contain at least three annotated subjects. Thus, the absolute Hit@1 values should not be read as a comparison across different subsets. Each row should be read as the D2 comparison between V_i and $\widetilde{V}_i^b$ within the same subset.

Under this rating, for Qwen3.5-9B, InternVL3.5-8B, and Ministral 3 8B, the $\widetilde{V}_i^b - V_i$ change moves from -45.8, -30.0, and -34.3 points in the single-subject group to -10.4, -3.7, and -7.0 points in the group with at least three annotated subjects. The subject-removal effect is therefore strongest when no other annotated subject remains visible. This stratification supports compensation by remaining person-level visual evidence, not scene-context substitution. AffectGPT remains nearly flat across all strata, matching its concentrated-output behavior in Table S7.

Moreover, for CAER, Table S12 repeats D2 removal with a localized pseudo face/head box. **On the retained clips, the median area of the box/marker changes from** 53.24% **of the frame for the pseudo person box to** 5.96% **for the pseudo face/head box. Due to the strict face/head-box generation process, around 50 samples are excluded (see SM G.1.2). The normal and control cases in Tables S12 and S13 therefore use the same filtered samples.** For the five general VLMs, masking the localized face/head region changes Acc@1 by -9.0 to -17.7 points. The corresponding pseudo person-box changes are -6.8 to -19.0 points on the same retained records. After face/head masking, the most frequent prediction is *Neutral* for all five general VLMs, with top-label shares from 57.8% to 96.1%. **The prediction collapse pattern therefore remains when the manipulated CAER region is localized to the face/head.**

E.2 Diagnosis 3: Subject–Scene Mismatch Controls

D3 controlled experiments focus on whether subject-region replacement (S_j, C_i) remains the stronger intervention than scene-context replacement (S_i, C_j). In EMOTIC, the marker result in Table S10 removes the explicit visual marker from (S_j, C_i). This changes Hit@1 by $+1.3$, $+5.0$, and $+1.6$ points for Qwen3.5-9B, InternVL3.5-8B, and Ministral 3 8B. These changes are small relative to the subject-region replacement drops in Table S9, so the D3 effect is not explained by the marker.

For CAER, Table S13 repeats both crossed inputs with the localized pseudo face/head box. For the five general VLMs, replacing the localized face/head region in (S_j, C_i) changes Acc@1 by -22.6 to -36.1 points, whereas replacing the remaining visual context in (S_i, C_j) changes Acc@1 by -2.8 to -5.8 points. The prediction-change rates follow the same ordering: 57.7%–79.9% for (S_j, C_i) and 26.2%–35.4%

TABLE S7

PREDICTION-DISTRIBUTION SUMMARIES FOR EVERY VISUAL INPUT ON THE VALIDATION AND TEST SPLITS, WITH ANNOTATION-FREQUENCY BASELINES. TOP-1 LABEL SHARE AND TOP-3 SHARE ARE PERCENTAGES OF THE MOST FREQUENTLY PREDICTED EMOTIONS AMONG ANSWERED SAMPLES. FOR ANNOTATION-FREQUENCY BASELINES, PARENTHESIZED VALUES DENOTE GROUND-TRUTH LABEL PREVALENCE FOR EMOTIC AND CLIP-LABEL FREQUENCY FOR CAER. N_{eff} IS THE EXPONENTIAL OF EMPIRICAL PREDICTION ENTROPY; DISTINCT LABELS COUNTS LABELS EMITTED AT LEAST ONCE.

Visual input / baseline	Statistic	Qwen3.5 -9B	InternVL3.5 -8B	Ministral 3 8B	Emotion -LLaMA	AffectGPT	GPT-5.6 Luna	Gemini-3.5 Flash-Lite
EMOTIC								
Annotation baseline	Top-1 annotated label	Engagement (82.7)						
Annotation baseline	Top-3 annotated labels	Engagement (82.7), Anticipation (60.2), Happiness (54.5)						
$V_i = (S_i, C_i)$	Top label share	Happiness (30.7)	Engagement (41.7)	Engagement (32.1)	Affection (81.4)	Happiness (93.6)	Engagement (43.9)	Engagement (64.5)
	Top-3 share	75.2	76.8	66.2	98.7	99.5	72.8	80.5
	N_{eff}	7.1	6.0	7.8	1.9	1.3	6.9	4.4
	Distinct labels	23	22	24	11	10	25	24
$\widetilde{V}_i^b$	Top label share	Disconnection (38.7)	Engagement (44.5)	Disconnection (29.7)	Affection (71.1)	Happiness (96.6)	Disconnection (60.4)	Engagement (69.9)
	Top-3 share	77.4	80.1	72.4	99.6	99.6	88.6	81.0
	N_{eff}	5.6	5.4	7.1	2.0	1.2	3.7	3.8
	Distinct labels	21	22	24	8	7	22	23
(S_i, C_j)	Top label share	Happiness (34.6)	Engagement (37.2)	Excitement (26.1)	Affection (89.2)	Happiness (92.8)	Engagement (33.0)	Engagement (61.7)
	Top-3 share	63.2	68.9	61.6	99.5	99.2	68.5	77.0
	N_{eff}	8.4	6.9	8.9	1.5	1.4	8.7	5.0
	Distinct labels	22	23	25	9	8	24	25
(S_j, C_i)	Top label share	Happiness (33.7)	Engagement (26.0)	Excitement (22.6)	Affection (88.5)	Happiness (93.1)	Happiness (27.2)	Engagement (40.0)
	Top-3 share	55.7	58.7	59.2	99.5	99.2	55.4	61.8
	N_{eff}	10.1	8.5	9.2	1.5	1.4	11.6	9.1
	Distinct labels	23	20	24	9	10	24	25
CAER								
Annotation baseline	Top-1 annotated label	Neutral (34.7)						
Annotation baseline	Top-3 annotated labels	Neutral (34.7), Happy (20.6), Anger (12.3)						
$V_i = (S_i, C_i)$	Top label share	Neutral (45.3)	Happy (43.6)	Neutral (45.8)	Surprise (67.0)	Neutral (98.2)	Neutral (42.2)	Neutral (55.5)
	Top-3 share	90.9	81.1	84.0	99.2	100.0	80.3	85.5
	N_{eff}	3.7	4.1	3.7	2.4	1.1	4.6	3.7
	Distinct labels	7	7	6	6	3	7	7
$\widetilde{V}_i^b$	Top label share	Neutral (95.2)	Neutral (85.9)	Neutral (94.2)	Anger (50.8)	Neutral (99.1)	Neutral (98.1)	Neutral (93.8)
	Top-3 share	99.5	98.5	98.6	99.7	100.0	99.5	98.2
	N_{eff}	1.3	1.7	1.3	2.8	1.1	1.1	1.4
	Distinct labels	6	6	7	4	2	6	7
(S_i, C_j)	Top label share	Neutral (43.1)	Happy (42.4)	Neutral (41.4)	Surprise (63.2)	Neutral (97.8)	Neutral (39.5)	Neutral (58.2)
	Top-3 share	89.2	82.0	85.0	99.1	100.0	77.5	85.6
	N_{eff}	3.9	4.2	3.9	2.5	1.1	4.9	3.6
	Distinct labels	7	7	6	5	4	7	7
(S_j, C_i)	Top label share	Neutral (39.4)	Happy (36.7)	Neutral (39.2)	Surprise (62.8)	Neutral (97.6)	Neutral (34.5)	Neutral (51.6)
	Top-3 share	86.6	84.9	80.9	99.1	100.0	73.3	81.3
	N_{eff}	4.1	4.2	4.0	2.5	1.1	5.2	4.0
	Distinct labels	7	7	6	5	4	7	7

for (S_i, C_j). **Thus, subject-region replacement remains stronger than scene-context replacement even when the CAER subject region is localized to the face/head**. AffectGPT and Emotion-LLaMA are included in the tables as affective VLMs, but they are not included in the numeric ranges above. AffectGPT's predictions are highly concen-

TABLE S8

PREDICTION-CHANGE RATES FOR D2 AND D3 ON THE VALIDATION AND TEST SPLITS. VALUES ARE **PERCENTAGES** OF PAIRS WHOSE PARSED PREDICTION DIFFERS FROM V_i. THE THREE COLUMNS IN EACH DATASET CORRESPOND TO V_i VERSUS $\widetilde{V}_i^b$, V_i VERSUS (S_i, C_j), AND V_i VERSUS (S_j, C_i), RESPECTIVELY.

Model	**EMOTIC**			**CAER**		
	$\widetilde{V}_i^b$	(S_i, C_j)	(S_j, C_i)	$\widetilde{V}_i^b$	(S_i, C_j)	(S_j, C_i)
GPT-5.6 Luna	74.9%	45.4%	87.6%	57.8%	26.2%	79.4%
Gemini-3.5 Flash-Lite	41.6%	30.1%	72.5%	45.9%	20.2%	69.1%
Qwen3.5-9B	61.3%	59.7%	81.0%	53.8%	16.9%	73.2%
InternVL3.5-8B	49.9%	48.6%	80.4%	80.4%	20.5%	77.6%
Ministral 3 8B	55.9%	68.3%	71.8%	54.3%	27.9%	72.3%
Emotion-LLaMA	23.4%	21.9%	23.4%	71.6%	24.7%	48.5%
AffectGPT	8.8%	12.4%	11.8%	2.5%	3.4%	4.1%

TABLE S9

RECOGNITION RESULTS FOR D2 AND D3. VALUES ARE PROPORTIONS, WITH PAIRED CHANGES FROM V_i IN PARENTHESES. V_i, $\widetilde{V}_i^b$, (S_i, C_j), AND (S_j, C_i) DENOTE THE ORIGINAL VISUAL INPUT, SUBJECT-REMOVED INPUT, SCENE-CONTEXT REPLACEMENT, AND SUBJECT-REGION REPLACEMENT, RESPECTIVELY.

Model	**Metric**	**EMOTIC**				**CAER**			
		V_i	$\widetilde{V}_i^b$	(S_i, C_j)	(S_j, C_i)	V_i	$\widetilde{V}_i^b$	(S_i, C_j)	(S_j, C_i)
GPT-5.6 Luna	Hit@1 / Acc@1	0.819	0.437(-0.381)	0.670(-0.148)	0.225(-0.593)	0.528	0.351(-0.177)	0.501(-0.026)	0.146(-0.382)
	Macro-F1	0.148	0.060(-0.089)	0.110(-0.038)	0.032(-0.116)	0.379	0.086(-0.293)	0.358(-0.021)	0.084(-0.295)
Gemini-3.5 Flash-Lite	Hit@1 / Acc@1	0.868	0.774(-0.094)	0.817(-0.051)	0.353(-0.515)	0.540	0.351(-0.189)	0.532(-0.007)	0.191(-0.349)
	Macro-F1	0.141	0.091(-0.050)	0.122(-0.019)	0.031(-0.110)	0.372	0.100(-0.273)	0.362(-0.011)	0.079(-0.293)
Qwen3.5-9B	Hit@1 / Acc@1	0.755	0.487(-0.269)	0.493(-0.262)	0.246(-0.509)	0.530	0.355(-0.175)	0.513(-0.017)	0.180(-0.350)
	Macro-F1	0.143	0.080(-0.062)	0.090(-0.052)	0.043(-0.100)	0.330	0.098(-0.233)	0.323(-0.007)	0.087(-0.243)
InternVL3.5-8B	Hit@1 / Acc@1	0.842	0.695(-0.147)	0.675(-0.166)	0.324(-0.518)	0.416	0.346(-0.069)	0.412(-0.004)	0.104(-0.311)
	Macro-F1	0.128	0.084(-0.044)	0.092(-0.036)	0.038(-0.091)	0.279	0.116(-0.163)	0.273(-0.006)	0.068(-0.210)
Ministral 3 8B	Hit@1 / Acc@1	0.754	0.563(-0.191)	0.513(-0.240)	0.385(-0.369)	0.456	0.344(-0.112)	0.445(-0.011)	0.187(-0.268)
	Macro-F1	0.132	0.083(-0.049)	0.077(-0.055)	0.057(-0.075)	0.275	0.095(-0.180)	0.265(-0.010)	0.091(-0.183)
Emotion-LLaMA	Hit@1 / Acc@1	0.187	0.156(-0.030)	0.157(-0.030)	0.144(-0.043)	0.220	0.127(-0.093)	0.219(-0.001)	0.107(-0.113)
	Macro-F1	0.029	0.019(-0.010)	0.019(-0.011)	0.016(-0.013)	0.152	0.079(-0.073)	0.151(-0.001)	0.051(-0.100)
AffectGPT	Hit@1 / Acc@1	0.553	0.550(-0.003)	0.559(+0.006)	0.549(-0.004)	0.347	0.344(-0.004)	0.348(+0.001)	0.343(-0.004)
	Macro-F1	0.032	0.029(-0.002)	0.032(+0.000)	0.031(-0.001)	0.081	0.075(-0.006)	0.085(+0.004)	0.080(-0.001)

TABLE S10

EMOTIC MARKER AND GRAY-RECTANGLE CONTROLS FOR D2 AND D3. ENTRIES ARE PAIRED HIT@1 CHANGES IN PERCENTAGE POINTS. FOR MARKER RESULTS, VALUES ARE MARKER-FREE MINUS MARKED. FOR GRAY-RECTANGLE RESULTS, $V_i^{\bar{b}_i}$ DENOTES THE INPUT WITH THE SUBJECT VISIBLE AND THE GRAY RECTANGLE PLACED OUTSIDE THE SUBJECT REGION; THE FIRST GRAY-RECTANGLE ROW REPORTS $V_i^{\bar{b}_i} - V_i$, AND THE SECOND REPORTS $V_i^{\bar{b}_i} - \widetilde{V}_i^b$.

Control comparison	**n**	**Qwen3.5-9B**	**InternVL3.5-8B**	**Ministral 3 8B**	**Emotion-LLaMA**	**AffectGPT**
Single annotated subject						
Remove marker from V_i	3905	+0.4	+0.3	+0.0	+0.2	−0.7
Remove marker from $\widetilde{V}_i^b$	3905	+7.1	+7.5	+1.7	+0.9	−0.3
Remove marker from (S_j, C_i)	3905	+1.3	+5.0	+1.6	−0.1	−0.3
Equal-area feasible subset						
Add gray rectangle away from subject to V_i	2739	+0.0	−0.2	−0.6	−0.8	+0.5
Use $V_i^{\bar{b}_i}$ instead of $\widetilde{V}_i^b$	2739	+41.4	+24.4	+29.3	+2.3	+1.0

trated on CAER, and Emotion-LLaMA remains below the CAER constant-label baseline. These patterns are consistent with the model-specific behavior reported in Table S7.

E.3 Mask Appearance and Resizing

We test how mask appearance and resizing affect the D2/D3 results using Qwen3.5-9B, InternVL3.5-8B, and

TABLE S11
EMOTIC D2 SUBJECT-REMOVAL RESULTS STRATIFIED BY THE NUMBER OF ANNOTATED SUBJECTS IN THE INPUT IMAGE. V_i AND $\widetilde{V}_i^b$ ENTRIES ARE HIT@1 PERCENTAGES, AND $\Delta = \widetilde{V}_i^b - V_i$ IS REPORTED IN PERCENTAGE POINTS.

Model	Annotated subjects	n	V_i Hit@1	$\widetilde{V}_i^b$ Hit@1	Δ
Qwen3.5-9B	1	3906	77.0	31.3	−45.8
-	2	3350	76.5	55.2	−21.3
-	≥ 3	3357	72.8	62.4	−10.4
InternVL3.5-8B	1	3906	86.0	56.0	−30.0
-	2	3350	83.7	75.7	−7.9
-	≥ 3	3357	82.6	78.9	−3.7
Ministral 3 8B	1	3906	79.4	45.2	−34.3
-	2	3350	74.8	61.2	−13.6
-	≥ 3	3357	71.3	64.3	−7.0
Emotion-LLaMA	1	3905	15.6	10.6	−5.0
-	2	3351	23.0	19.6	−3.4
-	≥ 3	3357	17.9	17.5	−0.4
AffectGPT	1	3906	50.0	49.3	−0.7
-	2	3350	57.4	57.3	−0.1
-	≥ 3	3357	59.3	59.4	+0.1

Ministral 3 8B.

Samples and evaluation. We use 350 CAER test clips, with 50 per emotion category, and 400 EMOTIC test images. CAER clips have face/head boxes and eye–nose landmarks in at least two frames. EMOTIC has 50 images in each of eight groups defined by single- versus multiple-person images and subject-area quartiles, with one annotated person selected per image. Results average these classes or groups equally. Prediction changes are measured relative to V_i; recognition uses Hit@1 on EMOTIC and Acc@1 on CAER.

Mask appearance. We replace the pixels inside the subject box with mid-gray or random RGB texture, keeping the box and marker fixed. Texture values are sampled independently from 0–255. Both fills reduce recognition for every model and dataset (Table S14(a)). Texture masking reduces Hit@1 by 20.75–31.50 percentage points on EMOTIC and Acc@1 by 13.43–18.86 points on CAER.

On CAER, Neutral accounts for 88.86–96.00% of predictions after gray masking and 88.00–98.57% after texture masking; Macro-F1 also drops under both fills (Table S14(b)). Prediction collapse therefore persists across mask appearances.

Resizing without changing the person. We resize the original subject crop to the paired sample's box size and back using bicubic interpolation, then return it to its original location. This keeps the person, scene, and marker unchanged.

Resizing changes 3.14–16.50% of predictions, compared with 72.25–83.00% for subject replacement on the same samples (Table S15). Recognition changes by -0.75 to $+0.29$ percentage points. Thus, resizing alone has a much smaller effect than replacing the subject.

Replacement with limited resizing distortion. To complement the resizing-only experiment, we test whether subject dominance persists when subject–scene replacements cause little geometric distortion. We use the existing validation and test predictions from all seven models, retaining samples with valid predictions for both original inputs and both recombinations. This compares subject and scene replacement on the same samples.

Subject replacement changes predictions more often than scene replacement for all five general VLMs on both datasets (Table S16).

All five general VLMs also show subject dominance in replacements with little resizing distortion. Here, the horizontal and vertical resizing factors differ by at most a factor of 1.25 in both recombinations and all four CAER frames. AffectGPT remains close to zero, and Emotion-LLaMA shows a smaller EMOTIC difference than the general VLMs.

Summary Prediction collapse on CAER persists across mask appearances, and subject dominance persists across resizing analyses, reinforcing the central role of subject evidence in the evaluated general VLMs.

E.4 Emotion Labels and Subject–Scene Replacement

Using the 350 CAER test clips and three models from the masking experiment, we compare replacements from clips with the same or a different emotion label.

Choosing replacement samples. For each original CAER clip i, we select four other clips: two with the same emotion label as i and two with a different label. The labels describe the person in each clip. Using each selected clip j, we replace the person to form (S_j, C_i) and replace the scene to form (S_i, C_j).

We compare clips using seven standardized features: person aspect ratio and area fraction, face-to-person area ratio (all log-transformed), horizontal and vertical face position, horizontal eye–nose displacement relative to inter-eye distance, and eye-line roll. Each feature is the median across eligible frames. From the 50 clips closest to i in each label group, we choose two pairs, each containing one same-label clip and one different-label clip. For each pair, we minimize the sum of the distances from its two clips to i and the distance between them. The four selected clips are distinct.

Effect of the selected clip's label. After averaging the two replacements in each label group for each original clip, different-label subject replacement changes predictions 15.57, 9.43, and 14.29 percentage points more often than same-label replacement for Qwen, InternVL, and Ministral (Table S17). For scene replacement, the corresponding differences are $+1.86$, -2.29, and $+1.57$ points.

On clips with a correct original prediction, the difference is 27.61–34.70 points. Same-label subject replacement already changes 55.14–61.57% of predictions, while different-label replacement produces an additional increase. The response to subject replacement is therefore partly associated with the emotion label of the selected clip.

Summary On CAER, the stronger response to different-label than same-label subject replacements supports the role of affective evidence in the observed subject dominance.

E.5 Prompt and Subject-View Sensitivity Analysis

We examine how the D2/D3 findings vary with prompt wording and how recognition varies with the subject view. The analysis uses the same three models on 140 CAER clips

TABLE S12

CAER FACE/HEAD-REGION CONTROL FOR SUBJECT-REGION REMOVAL ON THE VALIDATION AND TEST SPLITS. THE SUBJECT COLUMNS USE THE DEFAULT PSEUDO SUBJECT BOX, WHILE THE FACE/HEAD COLUMNS USE THE PSEUDO FACE/HEAD BOX ON THE SAME SUCCESSFUL RECORDS. V_i ACC@1 IS MEASURED ON THE ORIGINAL VISUAL INPUT, $\widetilde{V}_i^b$ ACC@1 MASKS THE CORRESPONDING REGION, $\Delta = \widetilde{V}_i^b - V_i$ IS REPORTED IN PERCENTAGE POINTS, AND TOP LABEL GIVES THE MOST FREQUENT PREDICTION AFTER REGION REMOVAL.

Model	subject region				Face/head region			
	V_i Acc@1	$\widetilde{V}_i^b$ Acc@1	Δ	Top label	V_i Acc@1	$\widetilde{V}_i^b$ Acc@1	Δ	Top label
GPT-5.6 Luna	52.84%	35.12%	-17.73	Neutral 98.13%	51.26%	35.17%	-16.09	Neutral 90.22%
Gemini-3.5 Flash-Lite	54.15%	35.18%	-18.99	Neutral 93.95%	53.79%	36.05%	-17.73	Neutral 88.34%
Qwen3.5-9B	53.05%	35.48%	-17.57	Neutral 95.26%	51.90%	35.37%	-16.52	Neutral 96.06%
InternVL3.5-8B	41.55%	34.73%	-6.81	Neutral 86.22%	42.19%	33.17%	-9.02	Neutral 57.81%
Ministral 3 8B	45.57%	34.40%	-11.17	Neutral 94.29%	46.26%	33.25%	-13.01	Neutral 86.14%
AffectGPT	34.78%	34.45%	-0.33	Neutral 99.15%	35.63%	34.73%	-0.90	Neutral 99.77%
Emotion-LLaMA	22.08%	12.78%	-9.30	Anger 50.92%	20.18%	12.04%	-8.15	Surprise 83.73%

TABLE S13

CAER FACE/HEAD-REGION CONTROL FOR EVIDENCE SWAPPING ON THE VALIDATION AND TEST SPLITS. SCENE-CONTEXT REPLACEMENT IS (S_i, C_j) AND SUBJECT-REGION REPLACEMENT IS (S_j, C_i). THE SUBJECT COLUMNS USE THE DEFAULT PSEUDO SUBJECT BOX, WHILE THE FACE/HEAD COLUMNS USE THE PSEUDO FACE/HEAD BOX ON THE SAME SUCCESSFUL PAIRED RECORDS. Δ ACC@1 IS THE CHANGE FROM THE CORRESPONDING ORIGINAL VISUAL INPUT IN PERCENTAGE POINTS, AND CHANGE IS THE PREDICTION-CHANGE RATE.

Model	Subject region				Face/head region			
	(S_i, C_j) Δ Acc@1	(S_i, C_j) Change	(S_j, C_i) Δ Acc@1	(S_j, C_i) Change	(S_i, C_j) Δ Acc@1	(S_i, C_j) Change	(S_j, C_i) Δ Acc@1	(S_j, C_i) Change
GPT-5.6 Luna	-2.59	26.15%	-38.25	79.40%	-5.80	35.35%	-36.03	79.89%
Gemini-3.5 Flash-Lite	-0.83	20.08%	-34.89	69.00%	-4.22	30.03%	-36.12	72.97%
Qwen3.5-9B	-1.74	16.79%	-34.91	73.21%	-3.01	26.22%	-32.81	71.00%
InternVL3.5-8B	-0.39	20.52%	-31.17	77.79%	-2.82	31.23%	-30.86	76.06%
Ministral 3 8B	-0.91	27.57%	-26.85	72.35%	-3.81	33.30%	-22.65	57.71%
AffectGPT	+0.10	3.45%	-0.39	4.15%	-0.34	2.82%	-1.19	3.58%
Emotion-LLaMA	-0.16	24.49%	-11.45	48.67%	-1.94	34.36%	-8.71	33.61%

TABLE S14

MASK-APPEARANCE SENSITIVITY. (A) RECOGNITION CHANGES UNDER GRAY AND RANDOM-TEXTURE SUBJECT REMOVAL (HIT@1 ON EMOTIC AND ACC@1 ON CAER). (B) NEUTRAL PREDICTIONS AND MACRO-F1 ON CAER.

(a) Recognition changes

Dataset	Model	Gray minus V_i (pp)	Texture minus V_i (pp)	Texture minus gray (pp)
EMOTIC	Qwen3.5-9B	−32.75	−31.50	+1.25
-	InternVL3.5-8B	−18.25	−20.75	−2.50
-	Ministral 3 8B	−20.00	−23.50	−3.50
CAER	Qwen3.5-9B	−18.57	−18.86	−0.29
-	InternVL3.5-8B	−13.43	−13.43	+0.00
-	Ministral 3 8B	−16.00	−15.14	+0.86

(b) Prediction concentration on CAER

Model	Neutral predictions (%)			Macro-F1		
	Original	Gray	Texture	Original	Gray	Texture
Qwen3.5-9B	45.14	95.43	94.57	0.2609	0.0559	0.0547
InternVL3.5-8B	15.14	88.86	88.00	0.2238	0.0733	0.0738
Ministral 3 8B	46.00	96.00	98.57	0.2299	0.0494	0.0472

and 160 EMOTIC images, with 20 samples per class or sampling group from Sec. E.3.

Prompt sensitivity. We keep the visual inputs, label set, and response format fixed across four prompts: the original

TABLE S15

RESAMPLING THE ORIGINAL SUBJECT REGION THROUGH THE PAIRED BOX SIZE AND BACK. CHANGE REPORTS HOW OFTEN THE PREDICTED LABEL DIFFERS FROM THAT FOR V_i. ANNOTATION-MATCH CHANGE IS THE DIFFERENCE IN HIT@1 ON EMOTIC OR ACC@1 ON CAER RELATIVE TO V_i. THE FINAL COLUMNS COMPARE SCENE AND SUBJECT REPLACEMENT ON THE SAME SAMPLES.

Dataset	Model	Resampling Change (%)	Annotation-match Change (pp)	(S_i, C_j) Change (%)	(S_j, C_i) Change (%)
EMOTIC	Qwen3.5-9B	16.00	−0.50	61.00	83.00
-	InternVL3.5-8B	9.50	−0.75	46.00	80.75
-	Ministral 3 8B	16.50	+0.25	66.50	72.25
CAER	Qwen3.5-9B	6.29	−0.57	18.57	76.00
-	InternVL3.5-8B	3.14	+0.29	20.29	76.86
-	Ministral 3 8B	7.43	+0.00	24.57	72.29

TABLE S16

SUBJECT VERSUS SCENE REPLACEMENT ON THE VALIDATION AND TEST SETS. VALUES REPORT HOW MUCH MORE OFTEN PREDICTIONS CHANGE FROM V_i AFTER SUBJECT REPLACEMENT THAN AFTER SCENE REPLACEMENT. SMALL RESIZING DISTORTION MEANS THAT THE HORIZONTAL AND VERTICAL RESIZING FACTORS DIFFER BY AT MOST A FACTOR OF 1.25 IN BOTH RECOMBINATIONS AND ALL FOUR CAER FRAMES.

Dataset	Model	All replacements (pp)	Small resizing distortion (pp)
EMOTIC	GPT-5.6 Luna	+42.20	+46.46
-	Gemini-3.5 Flash-Lite	+42.32	+43.23
-	Qwen3.5-9B	+21.38	+21.87
-	InternVL3.5-8B	+31.76	+34.50
-	Ministral 3 8B	+3.44	+9.56
-	Emotion-LLaMA	+1.51	+3.17
-	AffectGPT	−0.63	−0.73
CAER	GPT-5.6 Luna	+53.25	+55.58
-	Gemini-3.5 Flash-Lite	+48.91	+47.23
-	Qwen3.5-9B	+56.31	+56.80
-	InternVL3.5-8B	+57.10	+56.28
-	Ministral 3 8B	+44.40	+48.73
-	Emotion-LLaMA	+23.80	+28.18
-	AffectGPT	+0.71	+0.61

TABLE S17

SUBJECT AND SCENE REPLACEMENT ON CAER. *Same* AND *Different* INDICATE WHETHER THE SELECTED CLIP'S ANNOTATED EMOTION LABEL MATCHES THAT OF THE ORIGINAL CLIP. VALUES REPORT HOW OFTEN PREDICTIONS DIFFER FROM THOSE FOR V_i. *All* MEANS ALL CLIPS SELECTED FOR THIS EXPERIMENT ARE USED.*Correct* ON V_i MEANS CLIPS WHOSE ORIGINAL PREDICTION IS CORRECT ARE USED.

Model	Clips	Replacement	Same (%)	Different (%)	Different − same (pp)
Qwen3.5-9B	All	Subject	55.14	70.71	+15.57
Qwen3.5-9B	-	Scene	16.57	18.43	+1.86
InternVL3.5-8B	-	Subject	61.57	71.00	+9.43
InternVL3.5-8B	-	Scene	19.71	17.43	−2.29
Ministral 3 8B	-	Subject	56.00	70.29	+14.29
Ministral 3 8B	-	Scene	17.86	19.43	+1.57
Qwen3.5-9B	Correct on V_i	Subject	59.78	87.39	+27.61
Qwen3.5-9B	-	Scene	20.36	19.39	−0.97
InternVL3.5-8B	-	Subject	51.10	79.31	+28.21
InternVL3.5-8B	-	Scene	11.58	17.20	+5.61
Ministral 3 8B	-	Subject	47.72	82.42	+34.70
Ministral 3 8B	-	Scene	13.94	16.63	+2.69

prompt, two rewordings of its task sentence, and a prompt emphasizing the person's facial expression and body cues. Each altered input is compared with V_i under the same prompt. The prompt variants are given in Sec. G.6.

For D2, subject removal reduces recognition under all four prompts for every model and dataset (Table S18). The recognition loss persists under both rewording and explicit person emphasis.

For D3, subject replacement changes predictions more often than scene replacement for each model under all four prompts on both datasets (Table S18). On EMOTIC, Ministral shows smaller differences of 5.00–11.88 percentage points. Prompt wording affects the magnitude of this difference: person emphasis reduces InternVL's EMOTIC difference by 10.62 percentage points. The observed subject–scene ordering stays the same across prompts.

Subject-view sensitivity. We vary the subject view by presenting the original input, the subject crop alone, or both together. The combined input contains one image and its crop for EMOTIC, or four chronological frames followed by four corresponding crops for CAER. The combined-view prompt asks the model to use both person and scene cues. Recognition is compared on the same samples.

Cropping affects recognition differently across models and datasets (Table S19). With the crop alone, InternVL's CAER accuracy increases by 5.00 percentage points, while Qwen's EMOTIC Hit@1 decreases by 8.12 points. The other four settings also show decreases of 1.25–3.12 points. Adding the crop alongside the original input changes recognition by -0.62 to $+3.75$ points.

Summary Subject dominance persists across prompts while recognition varies with the subject view, reinforcing the need to evaluate visual grounding alongside recognition accuracy.

F. Internal States Analysis using Activation Patching

This section gives the internal-state counterpart to the PRMs introduced in Sec. 4.4. It tests whether visual-token states from the V_i forward pass can restore the full-input/original-input prediction in the subject-removed forward pass. We probe the internal states using **activation patching**.

F.1 Protocol

We use a fixed sample set of 768 EMOTIC samples: 256 from the validation split and 512 from the test split. For each split, we group samples by the number of ground-truth labels and by whether their labels include one of the five most frequent validation labels (*Engagement*, *Anticipation*, *Happiness*, *Excitement*, and *Confidence*). We use these five labels only to form a coarse frequent-versus-tail split without making the 768-sample subset too sparse. We then sample from these groups with seed 42, keeping the group proportions close to those in the split and preferring samples from different original images when possible. We only use a subset of the dataset because activation patching is substantially more expensive and requires repeated forward passes.

For each model, we first run both V_i and $\widetilde{V}_i^b$. The V_i pass provides the original-input prediction and the visual-token states to copy, while the $\widetilde{V}_i^b$ pass provides the subject-removed prediction to be patched. Restoration is then computed only on samples whose prediction changes after subject removal, $y_i^*(V_i) \neq y_i^*(\widetilde{V}_i^b)$. Samples without this change are excluded because there is no changed prediction to restore.

For each sample satisfying this condition, we then run $\widetilde{V}_i^b$ again while copying visual-token states from the V_i pass at one decoder depth. We run this three times: copying visual-token states whose image patches overlap the subject box b_i, copying visual-token states whose image patches do not overlap b_i, and copying all image visual-token states. The table names these three cases as subject visual tokens, other image visual tokens, and all image visual tokens. The reported restoration rate is the percentage of these samples whose patched prediction equals $y_i^*(V_i)$. Copying a pass into itself is used as an identity control and leaves predictions unchanged.

F.2 Choice of decoder depth

Table S20 compares restoration rates across decoder depths to determine the depth used in the subsequent analyses. At 25% depth, copying subject visual-token states results in the largest value in the Subject column for all three models, and copying all image visual-token states gives the largest value in the All image column for all three models. We therefore use 25% decoder depth for Table S21 and the PRMs.

At 75% depth, even copying all image visual-token states recovers the original-input prediction on only 8.2%, 6.3%, and 2.9% of these samples. The final layer gives zero recovery in the Subject, Other image, and All image columns for every model. We therefore do not use these later depths for the following analyses.

F.3 Restoration from Copied Visual-Token States

Table S21 reports restoration rates after copying subject visual-token states, other image visual-token states, and all image visual-token states at 25% decoder depth. The numbers of samples satisfying $y_i^*(V_i) \neq y_i^*(\widetilde{V}_i^b)$ are 377 for InternVL3.5-8B, 447 for Qwen3.5-9B, and 210 for Emotion-LLaMA, corresponding to 49%, 58%, and 27% of the fixed sample set.

The two general VLMs show that restoring subject-region-related states recovers the original-input prediction on substantially more samples than restoring states at other image positions. Besides, subject visual tokens occupy only 17.1% and 20.1% of image visual tokens, but restoring them recovers the original-input prediction on 60.5% and 79.6% of these samples. Restoring other image visual tokens recovers only 9.1% and 23.8%. **This contrast shows that restoring visual-token states inside the subject box has a much stronger effect than restoring visual-token states outside the subject box**.

The other-image column copies visual-token states outside the subject box. The scene is unchanged between V_i and $\widetilde{V}_i^b$; only the subject-box region is covered. The input-level subject-versus-scene comparison is provided by D3, where subject and scene evidence are both varied. Together with D3, this supports that **subject evidence plays a stronger role than scene context in the tested emotion predictions**.

TABLE S18

PROMPT SENSITIVITY OF D2 AND D3 ON EMOTIC AND CAER. D2 PREDICTION CHANGE REPORTS HOW OFTEN LABELS DIFFER BETWEEN V_i AND $\widetilde{V}_i^b$. D2 RECOGNITION CHANGES USE HIT@1 ON EMOTIC AND ACC@1 ON CAER. D3 DIFFERENCE IS THE PREDICTION CHANGE UNDER SUBJECT REPLACEMENT MINUS THAT UNDER SCENE REPLACEMENT. ALL COMPARISONS USE EACH PROMPT'S OWN PREDICTION FOR V_i.

Model	Prompt	EMOTIC			CAER		
		D2 prediction change (%)	D2 ΔHit@1 (pp)	D3 difference (pp)	D2 prediction change (%)	D2 ΔAcc@1 (pp)	D3 difference (pp)
Qwen3.5-9B	Original	58.75	−29.38	+24.38	53.57	−17.86	+55.71
-	Paraphrase 1	61.25	−28.75	+27.50	47.86	−13.57	+52.86
-	Paraphrase 2	59.38	−30.62	+26.88	53.57	−19.29	+57.86
-	Person emphasis	53.12	−14.38	+21.25	58.57	−18.57	+55.71
InternVL3.5-8B	Original	59.38	−20.62	+33.13	84.29	−16.43	+54.29
-	Paraphrase 1	62.50	−25.63	+36.25	83.57	−17.86	+51.43
-	Paraphrase 2	56.25	−18.75	+33.75	84.29	−18.57	+53.57
-	Person emphasis	51.88	−13.75	+22.50	80.71	−18.57	+56.43
Ministral 3 8B	Original	58.75	−21.25	+7.50	56.43	−18.57	+53.57
-	Paraphrase 1	61.87	−28.12	+11.88	55.00	−17.14	+54.29
-	Paraphrase 2	57.50	−23.75	+5.00	60.71	−17.86	+49.29
-	Person emphasis	63.12	−22.50	+8.75	55.71	−17.86	+50.00

TABLE S19

SUBJECT-VIEW SENSITIVITY OF RECOGNITION: ORIGINAL VISUAL INPUT, SUBJECT CROP, AND BOTH VIEWS. RECOGNITION IS HIT@1 ON EMOTIC AND ACC@1 ON CAER. CHANGE FROM V_i COMPARES RECOGNITION WITH THE ORIGINAL VISUAL INPUT. MACRO-F1 USES THE ORIGINAL DATASET LABEL SPACE.

Dataset	Model	Visual evidence	Recognition (%)	Macro-F1	Change from V_i (pp)
EMOTIC	Qwen3.5-9B	V_i	75.00	0.1861	0.00
-	-	Subject crop	66.88	0.1401	−8.12
-	-	Original + crop	77.50	0.1638	+2.50
-	InternVL3.5-8B	V_i	79.38	0.1497	0.00
-	-	Subject crop	76.25	0.1424	−3.12
-	-	Original + crop	78.75	0.1522	−0.62
-	Ministral 3 8B	V_i	69.38	0.1364	0.00
-	-	Subject crop	68.12	0.1241	−1.25
-	-	Original + crop	73.12	0.1642	+3.75
CAER	Qwen3.5-9B	V_i	31.43	0.2486	0.00
-	-	Subject crop	28.57	0.2190	−2.86
-	-	Original + crop	31.43	0.2476	+0.00
-	InternVL3.5-8B	V_i	29.29	0.2149	0.00
-	-	Subject crop	34.29	0.2626	+5.00
-	-	Original + crop	32.86	0.2557	+3.57
-	Ministral 3 8B	V_i	30.71	0.2349	0.00
-	-	Subject crop	28.57	0.2233	−2.14
-	-	Original + crop	30.71	0.2403	+0.00

The three columns are separate interventions rather than additive components. The subject and other-image values sum below the all-image value for InternVL3.5-8B, but above it for Qwen3.5-9B and Emotion-LLaMA.

Emotion-LLaMA shows the opposite spatial pattern, with other image visual tokens (56.6%) above subject visual tokens (33.3%). Its EMOTIC recognition is far below the constant-label baseline, with predictions concentrated on a single low-frequency label, which suggests a model-specific pattern.

F.4 Prediction restoration maps

A prediction restoration map (PRM) divides the image into grid cells and repeats the same restoration one cell at a time, as shown in the case-study figure in the main paper. A cell value is the change in the probability of $y_i^*(V_i)$ after copying

TABLE S20

DEPTH PROFILE FOR SUBJECT REMOVAL ON THE FIXED EMOTIC SUBSET. THE SUBJECT, OTHER IMAGE, AND ALL IMAGE COLUMNS REPORT THE PERCENTAGE OF SAMPLES WHOSE PREDICTION MATCHES THE ORIGINAL-INPUT FORWARD PASS AFTER COPYING THE CORRESPONDING VISUAL-TOKEN STATES FROM THAT PASS.

Model	Depth	Subject	Other image	All image
InternVL3.5-8B	25%	60.5	9.1	98.1
$n = 377$	50%	44.0	16.6	90.5
	75%	2.1	4.3	8.2
	final	0.0	0.0	0.0
Qwen3.5-9B	25%	79.6	23.8	96.9
$n = 447$	50%	26.6	21.3	61.3
	75%	1.3	5.2	6.3
	final	0.0	0.0	0.0
Emotion-LLaMA	25%	33.3	56.6	79.1
$n = 210$	50%	14.3	19.5	31.4
	75%	3.8	2.0	2.9
	final	0.0	0.0	0.0

TABLE S21

RESTORATION AFTER COPYING DIFFERENT VISUAL-TOKEN STATES ON THE FIXED EMOTIC SAMPLE SET. **SUBJECT TOKEN SHARE** MEASURES HOW MANY IMAGE VISUAL TOKENS FALL INSIDE THE SUBJECT BOX b_i. THE COLUMNS **SUBJECT VISUAL TOKENS**, **OTHER IMAGE VISUAL TOKENS**, AND **ALL IMAGE VISUAL TOKENS** REPORT THE PERCENTAGE OF SAMPLES WHOSE PREDICTION MATCHES THE ORIGINAL-INPUT FORWARD PASS AFTER THE CORRESPONDING VISUAL-TOKEN STATES FROM THAT PASS ARE RESTORED AT 25% DECODER DEPTH.

Model	n	Subject token share	Subject visual tokens	Other image visual tokens	All image visual tokens
InternVL3.5-8B	377	17.1%	60.5%	9.1%	98.1%
Qwen3.5-9B	447	20.1%	79.6%	23.8%	96.9%
Emotion-LLaMA	210	21.1%	33.3%	56.6%	79.1%

the visual-token state associated with that cell from the V_i pass into the subject-removed pass.

These examples explain where the restoration in Table S21 appears in individual images. For InternVL3.5-8B, the high-value cells are inside the marked subject region in all three rows. For Qwen3.5-9B, the first and third rows show a strong subject-region cell, while the second row has weaker cell-level restoration. This matches the table-level result that, for the two general VLMs, restoring subject visual-token states is more effective than restoring visual-token states outside the subject box.

Emotion-LLaMA does not show the same pattern. In the first two rows, high-value cells also appear outside the subject box. In the third row, the high-value cells are mostly inside the box. This mixed pattern is consistent with Table S21, where Emotion-LLaMA has higher restoration from other image visual-token states than from subject visual-token states. Thus, the PRMs support the statement that the two general VLMs rely more on subject visual-token states, and they show why Emotion-LLaMA is treated separately in this mechanism analysis.

G. IMPLEMENTATION SETTING

G.1 Pseudo Bounding Box Generation for CAER

CAER does not provide official subject boxes for this protocol, so we derive pseudo subject regions for the target subject associated with the clip-level emotion label. We build two pseudo-box sets: a default subject box that corresponds to the EMOTIC subject-box setting, and a localized face/head box for the CAER control.

G.1.1 Pseudo Subject Bounding Box: The default CAER subject box/marker is instantiated as a pseudo subject box. Its pipeline has four steps. 1) We decode each video and run YOLOv8 person detection on every frame, using image size 640, confidence threshold 0.1, detector non-maximum suppression (NMS) IoU 0.7, and an additional person-only NMS IoU 0.5.

2) The remaining detections are linked within each video with BoT-SORT. Track fragments separated by at most 12 frames are merged when their boxes satisfy temporal, center-distance, and area-ratio consistency checks.

3) We select one target track for the clip. The automatic score favors tracks with longer temporal coverage, higher detection confidence, more central boxes, larger subject regions, and stronger face/person association. Ambiguous multi-person clips are reviewed by humans.

4) After the target track is fixed, every decoded frame is assigned one pseudo subject bounding box. Direct tracked detections are used whenever available. If the selected track is missed between two detections, the marker is linearly interpolated. If it is missed before the first or after the last selected-track detection, the nearest selected-track box is propagated. The four-frame CAER inputs use the boxes at the same temporal-bin-center frames used as visual input. After generation, 15% of samples are randomly selected for human review.

The generated pseudo subject boxes cover all CAER evaluation samples used by D2-D3. The four-frame inputs therefore contain 15812 sampled frames with boxes. On these sampled frames, the median pseudo subject-box area is 53.1% of the frame and the mean is 54.3%.

G.1.2 Pseudo Face/Head Bounding Box: For the CAER face/head-region control, the localized box is generated after the pseudo subject track has been selected. We reuse the selected target track, the temporal-bin-center frame-sampling parameters, and the prompt from the pseudo subject-box setting. Only the region used for S is changed from the default pseudo subject box to the localized pseudo face/head box.

On the same sampled frames, YuNet first detects face candidates. A face candidate is accepted only when it is consistent with the selected subject box, landmarks, visible face area, and temporal continuity. The main thresholds are face detection confidence 0.6, face NMS IoU 0.3, associated-face confidence 0.75, and face/subject overlap 0.6. When no reliable face is accepted, a head box is used only if the target head is localized within the selected subject box. Similar to

SM G.1.1, 15% of samples are randomly selected for human review.

For this control, a clip is retained only when all four sampled frames have a face or head region. Missing face/head boxes are not replaced with pseudo subject boxes. This keeps 3,904 of the 3,953 CAER clips (98.8%). The median face/head-box area is 5.9% of the frame, and the mean is 7.2%, which is much smaller than the pseudo subject-box area.

G.2 Prompt for Diagnosis 1

Basic prompt. Used for all models and all four D1 conditions (image present / blank / noise / blank+mirage-hint), and as the control prompt for C.2–C.4.

D1: Basic prompt

```
What emotion is the person in this image
expressing? Choose exactly one:
happy, sad, angry, fear, surprise,
disgust, neutral.
Respond with only the label as: [[label]].
```

Parsing accepts a single `[[·]]` span; the label is canonicalized to the seven-class taxonomy (*angry*→*anger*). A non-empty non-matching label maps to *other*; a missing span is a parsing failure.

Abstention-option prompt (AO, C.4). The basic prompt with *cannot determine* appended as an eighth option; all other text is unchanged. Only selecting *cannot determine* (as `[[cannot determine]]`, `[cannot determine]`, bare or bold text, case-insensitive) counts as an abstention. Closed-source models use the structured-output (JSON enum) version of the same option set.

D1: Abstention-option prompt (AO)

```
What emotion is the person in this image
expressing? Choose exactly one:
happy, sad, angry, fear, surprise,
disgust, neutral, cannot determine
Respond with only the label as: [[label]].
```

Open-ended prompt (C.1). Removes the fixed option menu and offers no abstention option; the model is asked the emotion question directly. Any named emotion counts as a mirage, whereas a spontaneous *"no person / cannot determine"* is typed as an abstention (the only non-MIRAGE outcome).

D1: Open-ended prompt (C.1)

```
What emotion is the person in this image
expressing?
Respond with only the label as: [[label]].
```

Label-order permutation (C.3). The option line "`happy, sad, angry, fear, surprise, disgust, neutral`" is replaced by each of five fixed permutations of the seven words; ordering 0 is the canonical control, and orderings 1–4 place *neutral* in the first, middle, last, and second positions to test whether the collapse target depends on option position. All other text is unchanged.

Blank+mirage-hint metadata. For the hint condition the blank image is accompanied by misleading metadata asserting that an image is attached; the prompt body is the basic prompt (or the AO prompt) above.

G.3 Prompt for Diagnosis 2

D2 uses no context block. Within each dataset, the same prompt is used for V_i and $\widetilde{V}_i^b$; only the visual input changes.

EMOTIC system prompt.

D2: EMOTIC system prompt

You are performing apparent emotion recognition for one designated target person.
Predict the single most applicable apparent emotion category for the designated target person based only on the information provided.
A visual input may or may not be provided. If a full image is provided, the designated target person is indicated by a thin red bounding box. If a cropped person image is provided, the person shown in the crop is the designated target person and is enclosed by a thin red bounding box. Do not invent or assume information that is not provided.
Choose exactly ONE emotion from the following fixed emotion label set, using the exact spelling: ["Affection", "Anger", "Annoyance", "Anticipation", "Aversion", "Confidence", "Disapproval", "Disconnection", "Disquietment", "Doubt/Confusion", "Embarrassment", "Engagement", "Esteem", "Excitement", "Fatigue", "Fear", "Happiness", "Pain", "Peace", "Pleasure", "Sadness", "Sensitivity", "Suffering", "Surprise", "Sympathy", "Yearning"].
Return only one valid JSON object. It must contain exactly one key named `predicted_emotions`. Its value must be a JSON array containing exactly one string selected from the label set above. Do not return any other keys or text.

The EMOTIC user prompt for these visual conditions is:

D2: EMOTIC user prompt

Target: The designated target person for this sample.
Visual input: Provided.

CAER system prompt.

D2: CAER system prompt

You are performing clip-level apparent emotion recognition for one designated target person.
Predict the single most applicable apparent emotion category for the designated target person based only on the information provided.
A visual input may or may not be provided. When provided, it consists of four silent frames from the same short video in chronological order. If full frames are provided, the same designated target person is indicated in every frame by a thin red bounding box. If cropped person frames are provided, the designated person fills each crop and is enclosed by a thin red bounding box. Do not invent or assume information that is not provided. No audio is available.
Choose exactly ONE emotion from the following fixed label set, using the exact spelling: ["Anger", "Disgust", "Fear", "Happy", "Neutral", "Sad", "Surprise"].
Return only one valid JSON object. It must contain exactly one key named `predicted_emotions`. Its value must be a JSON array containing exactly one string selected from the label set above. Do not return any other keys or text.

The CAER user prompt for these visual conditions is:

D2: CAER user prompt

Target: The designated target person across this short video clip.
Visual input: Four silent chronological frames provided.

G.4 Prompt for Diagnosis 3

D3 uses the D2 system and user prompts verbatim for both datasets. The prompt does not state whether the visual input is V_i, (S_i, C_j), or (S_j, C_i); the distinction is entirely in the constructed visual input.

G.5 Prompt for Diagnosis 4

Scoring prompt. D4 confidence is obtained by teacher-forced scoring (Eq. 18) rather than sampling. FER+ is evaluated over an eight-class taxonomy (the seven canonical classes plus *contempt*); the option word list is inserted verbatim into the frozen answer format:

D4: Scoring prompt

```
What emotion is the person in this image
expressing? Choose exactly one:
happy, sad, angry, fear, surprise,
disgust, contempt, neutral.
Respond with only the label as: [[label]].
```

For each class c we score the bracketed continuation `[[v]]` for every surface form $v \in \mathcal{V}_c$, marginalize over $\mathcal{V}_c$ by log-sum-exp, then softmax-normalize across the eight classes (Eq. 18). Entropy and Jensen–Shannon divergence are computed in base 2.

Verbalizer sets $\mathcal{V}_c$. Surface forms per class; only *anger* realizes two forms, since the anger token is model-dependent (some models emit `[[angry]]`, others `[[anger]]`), so both are marginalized:

Canonical class	Surface forms $\mathcal{V}_c$
neutral	neutral
happy	happy
sad	sad
anger	angry, anger
surprise	surprise
disgust	disgust
fear	fear
contempt	contempt

Note on model scope. Teacher-forced scoring requires access to token logits, so D4 confidence metrics are computed for the open models only; closed-source values (where reported) are reference-only because their output heads are inaccessible.

G.6 Full Prompts for Sensitivity Analysis

EMOTIC

EMOTIC: Paraphrase 1 (system)

You are performing apparent emotion recognition for one designated target person.
Using only the supplied visual information, select the one apparent emotion category that best describes the designated target person.
A visual input may or may not be provided.
If a full image is provided, the designated target person is indicated by a thin red bounding box.
If a cropped person image is provided, the person shown in the crop is the designated target person and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided.
Choose exactly ONE emotion from the following fixed emotion label set, using the exact spelling:
["Affection", "Anger", "Annoyance", "Anticipation", "Aversion", "Confidence", "Disapproval", "Disconnection", "Disquietment", "Doubt/Confusion", "Embarrassment", "Engagement", "Esteem", "Excitement", "Fatigue", "Fear", "Happiness", "Pain", "Peace", "Pleasure", "Sadness", "Sensitivity", "Suffering", "Surprise", "Sympathy", "Yearning"]
Return only one valid JSON object. It must contain exactly one key named "predicted_emotions". Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

EMOTIC: Paraphrase 1 (user)

Target:
The designated target person for this sample.
Visual input:
Provided.

EMOTIC: Paraphrase 2 (system)

You are performing apparent emotion recognition for one designated target person.
For the designated target person, identify the best-fitting apparent emotion from the listed categories using the provided evidence only.
A visual input may or may not be provided.
If a full image is provided, the designated target person is indicated by a thin red bounding box.
If a cropped person image is provided, the person shown in the crop is the designated target person and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided.
Choose exactly ONE emotion from the following fixed emotion label set, using the exact spelling:
["Affection", "Anger", "Annoyance", "Anticipation", "Aversion", "Confidence", "Disapproval", "Disconnection", "Disquietment", "Doubt/Confusion", "Embarrassment", "Engagement", "Esteem", "Excitement", "Fatigue", "Fear", "Happiness", "Pain", "Peace", "Pleasure", "Sadness", "Sensitivity", "Suffering", "Surprise", "Sympathy", "Yearning"]
Return only one valid JSON object. It must contain exactly one key named "predicted_emotions". Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

EMOTIC: Paraphrase 2 (user)

Target:
The designated target person for this sample.
Visual input:
Provided.

EMOTIC: Person emphasis (system)

You are performing apparent emotion recognition for one designated target person.
Predict the single most applicable apparent emotion category for the designated target person based only on the information provided.
A visual input may or may not be provided.
If a full image is provided, the designated target person is indicated by a thin red bounding box.
If a cropped person image is provided, the person shown in the crop is the designated target person and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided.
Choose exactly ONE emotion from the following fixed emotion label set, using the exact spelling:
[”Affection”, ”Anger”, ”Annoyance”, ”Anticipation”, ”Aversion”, ”Confidence”, ”Disapproval”, ”Disconnection”, ”Disquietment”, ”Doubt/Confusion”, ”Embarrassment”, ”Engagement”, ”Esteem”, ”Excitement”, ”Fatigue”, ”Fear”, ”Happiness”, ”Pain”, ”Peace”, ”Pleasure”, ”Sadness”, ”Sensitivity”, ”Suffering”, ”Surprise”, ”Sympathy”, ”Yearning”]
Return only one valid JSON object. It must contain exactly one key named ”predicted_emotions”. Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

EMOTIC: Person emphasis (user)

Target:
The designated target person for this sample.
Visual input:
Provided.
Prioritize the designated person’s visible facial expression and body cues; do not let the surrounding scene override those cues.

EMOTIC: Original + crop (system)

You are performing apparent emotion recognition for one designated target person.
Predict the single most applicable apparent emotion category for the designated target person based only on the information provided.
A visual input may or may not be provided.
If a full image is provided, the designated target person is indicated by a thin red bounding box.
If a cropped person image is provided, the person shown in the crop is the designated target person and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided.
Choose exactly ONE emotion from the following fixed emotion label set, using the exact spelling:
[”Affection”, ”Anger”, ”Annoyance”, ”Anticipation”, ”Aversion”, ”Confidence”, ”Disapproval”, ”Disconnection”, ”Disquietment”, ”Doubt/Confusion”, ”Embarrassment”, ”Engagement”, ”Esteem”, ”Excitement”, ”Fatigue”, ”Fear”, ”Happiness”, ”Pain”, ”Peace”, ”Pleasure”, ”Sadness”, ”Sensitivity”, ”Suffering”, ”Surprise”, ”Sympathy”, ”Yearning”]
Return only one valid JSON object. It must contain exactly one key named ”predicted_emotions”. Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

EMOTIC: Original + crop (user)

Visual input:
Image 1: the full image with the target marked in red. Image 2: a crop of that same target person.
Consider the visible person cues in the crop, then combine them with the scene information in the full image to choose the final category.

CAER

CAER: Paraphrase 1 (system)

You are performing clip-level apparent emotion recognition for one designated target person.
Using only the supplied visual information, select the one apparent emotion category that best describes the designated target person.
A visual input may or may not be provided. When provided, it consists of four silent frames from the same short video in chronological order.
If full frames are provided, the same designated target person is indicated in every frame by a thin red bounding box.
If cropped person frames are provided, the designated person fills each crop and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided. No audio is available.
Choose exactly ONE emotion from the following fixed label set, using the exact spelling:
[”Anger”, ”Disgust”, ”Fear”, ”Happy”, ”Neutral”, ”Sad”, ”Surprise”]
Return only one valid JSON object. It must contain exactly one key named ”predicted_emotions”. Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

CAER: Paraphrase 1 (user)

Target:
The designated target person across this short video clip.
Visual input:
Four silent chronological frames provided.

CAER: Paraphrase 2 (system)

You are performing clip-level apparent emotion recognition for one designated target person.
For the designated target person, identify the best-fitting apparent emotion from the listed categories using the provided evidence only.
A visual input may or may not be provided. When provided, it consists of four silent frames from the same short video in chronological order.
If full frames are provided, the same designated target person is indicated in every frame by a thin red bounding box.
If cropped person frames are provided, the designated person fills each crop and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided. No audio is available.
Choose exactly ONE emotion from the following fixed label set, using the exact spelling:
[”Anger”, ”Disgust”, ”Fear”, ”Happy”, ”Neutral”, ”Sad”, ”Surprise”]
Return only one valid JSON object. It must contain exactly one key named ”predicted_emotions”. Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

CAER: Paraphrase 2 (user)

Target:
The designated target person across this short video clip.
Visual input:
Four silent chronological frames provided.

CAER: Person emphasis (system)

You are performing clip-level apparent emotion recognition for one designated target person.
Predict the single most applicable apparent emotion category for the designated target person based only on the information provided.
A visual input may or may not be provided. When provided, it consists of four silent frames from the same short video in chronological order.
If full frames are provided, the same designated target person is indicated in every frame by a thin red bounding box.
If cropped person frames are provided, the designated person fills each crop and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided. No audio is available.
Choose exactly ONE emotion from the following fixed label set, using the exact spelling:
[”Anger”, ”Disgust”, ”Fear”, ”Happy”, ”Neutral”, ”Sad”, ”Surprise”]
Return only one valid JSON object. It must contain exactly one key named ”predicted_emotions”. Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

CAER: Person emphasis (user)

Target:
The designated target person across this short video clip.
Visual input:
Four silent chronological frames provided.
Prioritize the designated person’s visible facial expression and body cues; do not let the surrounding scene override those cues.

CAER: Original + crop (system)

You are performing clip-level apparent emotion recognition for one designated target person.
Predict the single most applicable apparent emotion category for the designated target person based only on the information provided.
A visual input may or may not be provided. When provided, the first four images are silent full frames in chronological order, and images 5–8 are crops of the designated person from those same frames in the same order.
If full frames are provided, the same designated target person is indicated in every frame by a thin red bounding box.
If cropped person frames are provided, the designated person fills each crop and is enclosed by a thin red bounding box.
Do not invent or assume information that is not provided. No audio is available.
Choose exactly ONE emotion from the following fixed label set, using the exact spelling:
[”Anger”, ”Disgust”, ”Fear”, ”Happy”, ”Neutral”, ”Sad”, ”Surprise”]
Return only one valid JSON object. It must contain exactly one key named ”predicted_emotions”. Its value must be a JSON array containing exactly one string selected from the label set above.
Do not return any other keys or text.

CAER: Original + crop (user)

Visual input:
Images 1–4: chronological full frames. Images 5–8: the corresponding target-person crops. No audio.
Consider the visible person cues in the crops, then combine them with the scene information in the full frames to choose the final category.